

Trabajo de Fin de Máster
Máster Universitario en Lógica, Computación e IA

A Study on Foundational Visual-Language
Models for Histopathology Image Analysis

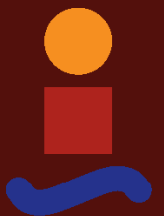
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Dpto. de Ciencias de la Computación e IA
Escuela Técnica Superior de Ingeniería Informática
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Image Analysis

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*Leandro Candau Sánchez de Ybargüen
Sevilla, 2026*

Resumen

- **Contexto.** Clasificación de imágenes histopatológicas bajo el cuello de botella recurrente del dominio: datos etiquetados limitados y vocabularios de etiquetas estrechos.
- **Enfoque.** Aprendizaje contrastivo imagen–texto al estilo CLIP con codificadores fundacionales congelados. Sobre una línea base controlada (ResNet50 + BERT congelados) se evalúa una matriz de ablaciones de una sola variable, entrenadas en Kaggle LC25000 (5 clases de tejido pulmonar y colónico) y evaluadas tanto sobre LC25000 como sobre tres conjuntos externos no vistos en entrenamiento.
- **Resultado principal.** El *preentrenamiento del codificador de imagen* es, con diferencia, la palanca dominante. Manteniendo el resto del pipeline idéntico y sustituyendo el ResNet50 de propósito general (ImageNet) por el ViT-B/32 preentrenado en histopatología de PLIP, la macro-F1 externa mejora en +0.067 en colon de cercanía (NCT-CRC), +0.027 en colon de otro hospital (Chaoyang) y +0.335 en pulmón de otro órgano (LungHist700).
- **Descomposición con DINOv2.** Un ViT-B/14 con preentrenamiento autosupervisado en imágenes naturales generales (sin histopatología) eleva LungHist700 en +0.152 sobre la línea base — aproximadamente la mitad de la ganancia de PLIP. La otra mitad procede del preentrenamiento específico en histopatología.
- **Hallazgos secundarios.** De las otras cinco intervenciones probadas, la normalización de tinción Macenko (referencia NCT-NORM) es la única palanca del lado de la entrada que produce una ganancia positiva medible bajo desplazamiento (+0.030 en Chaoyang, +0.020 en LungHist700), pero su magnitud queda dominada por la palanca del codificador de imagen en los mismos conjuntos y regresa en NCT-CRC (−0.022, que llega ya normalizado por Macenko en su *release*). Los codificadores de texto biomédicos, el clasificador de entropía cruzada sin CLIP y las variaciones de formato de *prompt* no mejoran la generalización. La ventaja de CLIP sobre un clasificador no-CLIP aparece sólo bajo desplazamiento más fuerte, vía la geometría *coseno + L2-normalizada + escalada por temperatura* del clasificador.
- **Aporte metodológico.** El reporte por órgano es esencial: agregar los conjuntos externos en un único número promedio oculta el cambio de régimen entre colon (transferencia parcial con ResNet50/ImageNet congelado) y pulmón (fallo categórico del mismo *backbone*).

Abstract

- **Context.** Histopathology image classification under the domain’s recurring bottleneck: limited labelled data and narrow label vocabularies.
- **Approach.** CLIP-style image–text contrastive learning with frozen foundation-model encoders. A single-variable ablation matrix is evaluated over a controlled baseline (frozen ResNet50 + frozen BERT), trained on Kaggle LC25000 (5 lung and colon tissue classes) and evaluated on both LC25000 and three additional external datasets the model has never seen.
- **Headline result.** The *image encoder’s pretraining* is by far the dominant lever. Holding the rest of the pipeline identical and swapping the general-purpose ResNet50 (ImageNet) for PLIP’s histopathology-pretrained ViT-B/32 lifts external macro-F1 by +0.067 on close-shift colon (NCT-CRC), +0.027 on different-hospital colon (Chaoyang), and +0.335 on different-source lung (LungHist700).
- **DINOv2 decomposition.** A ViT-B/14 with self-supervised pretraining on general natural images (no histopathology data) lifts LungHist700 by +0.152 over baseline — roughly half of PLIP’s gain. The other half comes from histopathology-specific pretraining.
- **Secondary findings.** Among the other five interventions tested, Macenko stain normalization (NCT-NORM reference) is the only input-side lever that produces a measurable positive gain under shift (+0.030 on Chaoyang, +0.020 on LungHist700), but its magnitude is dominated by the image-encoder lever on the same datasets and it regresses on NCT-CRC (−0.022, which arrives Macenko-normalised at release). Biomedical text encoders, a non-CLIP cross-entropy classifier, and prompt-format variations do not improve generalization. CLIP’s edge over a non-CLIP classifier appears only under heavier shift, via the *bounded-cosine + L2-normalised + temperature-scaled* classifier geometry.
- **Methodological contribution.** Per-organ external reporting is essential: aggregating external datasets into a single mean-OOD number hides the regime change between colon (partial transfer with a frozen ImageNet ResNet50) and lung (categorical failure of the same backbone).

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1 Introduction

Learning directly from raw text about images is a promising alternative which leverages a much broader source of supervision.

RADFORD ET AL., 2021

1.1 Motivation

- Clinical relevance of histopathology classification.
- Limited labelled data — a recurring limitation in this domain.
- Foundation models [1] — able to do different tasks with little task-specific supervision; in the vision–language line, CLIP [2] is the paradigm under study here.
- Project question (refined during the project): “*Where does the pretraining of a CLIP-style histopathology pipeline pay off, and where does it not?*” — evaluated across head-side, loss-side, text-side, and image-side interventions, both in-distribution and on three external datasets covering two organs.
- Framing: CLIP as the object of study, not the classifier. Findings should generalise to other scientific domains where short or missing labels are the norm.

1.2 Project Scope Evolution

- Initial direction → semantic segmentation. Abandoned because annotated segmentation data was too scarce within the time budget.
- Pivot → image-level classification, same domain, reframed question.
- Training source: LC25000 [3] (5-class, lung + colon).
- External evaluation: NCT-CRC-HE-7K [4] (close-shift colon), Chaoyang [5] (different-hospital colon), and LungHist700 [6] (different-source lung).
- Reference points: PLIP [7] zero-shot (no LC25000 supervision) and a plain image classifier (no CLIP architecture).

1.3 Research Questions

- **RQ1** — What baseline classification and generalization behaviour does a frozen general-purpose CLIP pipeline (ImageNet ResNet50 + BERT) achieve on LC25000, and what failure modes appear?

- **RQ2** — Does the CLIP paradigm validate its *foundational-model* premise — learn one task, generalize to tasks it was not specifically trained for — relative to a non-CLIP classifier on the same image encoder?
- **RQ3** — Does stain normalization help when we evaluate on datasets the model was never trained on?
- **RQ4** — Does using a text encoder pretrained on biomedical text (BioBERT [8], PubMedBERT [9]), instead of generic BERT [10], improve performance under short class-name prompts?
- **RQ5** — Does the image encoder’s *pretraining choice* matter? Specifically: replacing the general-purpose ResNet50 [11] with PLIP’s histopathology-pretrained ViT-B/32 [7, 12], and with DINOv2’s general-domain self-supervised ViT-B/14 [13] as a control that isolates “ViT architecture and self-supervision” from “histopathology pretraining specifically”.
- **RQ6** — How well does each variant generalize to data the model has never seen, evaluated across three external datasets that span shift severity and two organs?

1.4 Contributions

The central contribution of this project is a controlled measurement of how much each major design choice in a CLIP-style histopathology pipeline actually matters, with every claim framed as a delta over the same baseline — frozen ResNet50 + frozen BERT, single fixed prompt, stratified 80/10/10 split on LC25000 — and iterated through one experiment at a time in Chapter 4.

Headline findings (full results in Chapter 4):

- **Image encoder pretraining is the dominant lever.** Replacing ResNet50 with PLIP’s pretrained ViT-B/32 (histopathology image–text contrastive) gains +0.067 F1 on close-shift colon (NCT-CRC), +0.027 on different-hospital colon (Chaoyang), and +0.335 on different-source lung (LungHist700). No other intervention tested moves the needle comparably.
- **The lung-side advantage decomposes evenly.** The DINOv2 control (same general ViT architecture family as PLIP, self-supervised on general natural images, no histopathology data) lifts LungHist700 by +0.152 over baseline — about half of PLIP’s +0.335 gap. The other half is attributable to histopathology pretraining specifically.
- **Five other interventions move the needle far less, or only in narrow regimes.** Stain Macenko (NCT-NORM reference) is the only input-side lever that produces a measurable positive gain under shift (+0.030 on Chaoyang, +0.020 on LungHist700), but its magnitude is dominated by the image-encoder lever on the same datasets and it regresses on NCT-CRC (−0.022, which arrives Macenko-normalised at release). Biomedical text encoders (BioBERT, PubMedBERT) tie or trail baseline across all 6 grid cells on NCT, and the best biomedical cell (PubMedBERT + name_only) extended to Chaoyang and LungHist700 confirms no transfer benefit on any of the three datasets. A non-CLIP plain classifier ties CLIP in-distribution and on close-shift external data but loses on heavier shift; prompt-format variations within the text encoder ablation are within noise.
- **CLIP’s foundational-model premise is validated under shift, not in-distribution.** CLIP and a plain CE classifier tie at saturation in-distribution and on close-shift external; CLIP’s edge appears under heavier shift via the bounded-cosine + temperature-scaled classifier geometry, which doesn’t degenerate the way high-confidence softmax does.

Methodological contributions:

- Single-variable-change protocol: each experiment differs from the baseline in exactly one design choice, so the resulting delta is attributable.
- Per-organ external reporting: results are presented per dataset, not averaged; aggregate mean-OOD numbers would hide the cross-organ regime change observed in this study.
- Reproducibility: artifact bundle (fixed-seed split, cached DINOv2 features, weights, metric files, plots) versioned alongside the report in the project repository.
- Explicit written discussion of dataset and methodological limits (Chapter 5).

1.5 Document Structure

- Chapter 1 (Introduction) — motivation, scope, research questions, contributions.
- Chapter 2 (State of the Art) — foundation models, multimodal contrastive lineage, pathology FMs, where this thesis fits.
- Chapter 3 (Methodology) — CLIP architecture, image and text encoders, datasets, preprocessing, baseline, evaluation protocol, reproducibility.
- Chapter 4 (Results and Discussion) — the experiments and what each measured, ending in a synthesis section.
- Chapter 5 (Conclusions and Future Work) — one-paragraph-per-RQ conclusions, limitations, future work.

2 State of the Art

AI is undergoing a paradigm shift with the rise of models [...] that are trained on broad data at scale and are adaptable to a wide range of downstream tasks.

BOMMASANI ET AL., 2021

- Foundation models framing (Bommasani et al. [1], 2021).
- CLIP and multimodal contrastive learning (Radford et al. [2], 2021; see also ALIGN [14]) — the paradigm under study.
- Self-supervised visual pretraining — contrastive (SimCLR [15], MoCo [16]), self-distillation (DINO [17]), and scaled SSL on diverse natural images (DINOv2 [13]).
- Pathology shift: per-tile supervised CNNs → foundation models (2016–2024).
- Pathology-specific foundation models:
 - TransPath [18] (2022) — transformer + SSL.
 - Phikon [19] (2023) — masked image modelling.
 - PLIP [7] (2023) — CLIP on medical Twitter.
 - CONCH [20] (2024) — larger pathology VLM.
 - UNI [21] (2024), BiomedCLIP [22] (2023), MUSK [23] (2025) — recent entrants.

TODO before shipping — verify literature claim. Conclusion §5.1 (RQ5) asserts that “recent CLIP-for-histopathology papers routinely emphasise novel contrastive losses, biomedical text encoders, and stain harmonisation as the key levers for cross-dataset generalisation”. Before submission, skim the pathology foundation-model papers listed above (CONCH, UNI, BiomedCLIP, MUSK, PLIP, TransPath, Phikon) and confirm that this characterisation holds. If accurate, surface the relevant citations here and remove this note; if inaccurate, soften the Conclusion claim accordingly.

- Open challenges: benchmarks, dataset bias, cross-institution generalization, compute scale.
- Where this project fits: small-scale empirical study of general-purpose foundation backbones on pathology classification, with single-variable ablations.

3 Methodology

3.1 Overview

- One-page chapter map + project question recap.
- Reading order: background → baseline → ablations.

3.2 Histopathology Digital Imaging

- H&E staining.
- Whole-Slide Imaging + patch extraction.
- Stain variability and normalization (Macenko [24], Reinhard [25], Vahadane [26]).
- Why the domain is hard for general-purpose VLMs: domain gap from natural images, no per-image captions in clinical archives, dataset artifacts that can drive performance.

3.3 Contrastive Vision-Language Learning (CLIP)

3.3.1 Why CLIP?

- CLIP as object of study, not the classifier.
- Capabilities CLIP provides over a softmax head:
 - Open-vocabulary classification (new classes via prompts proven by adding new samples).
 - Semantic similarity output (cosine vs natural language).
 - Prompt engineering as a sensitivity lever.

3.3.2 Architecture and training signal

- Dual-encoder, shared d -dimensional embedding space ($d = 256$ throughout this thesis).
- Symmetric InfoNCE loss with a learnable temperature τ (formal definition below).
- In-batch false-negative regime at small batch + few classes (discussed under *Caveats* below).
- Prompt-based zero-shot classification at inference: an image is classified by argmax of cosine similarity against the encoded class prompts.
- CLIP-paper sample-level retrieval → doesn't apply to our setup: every image in a class shares the same prompt, so retrieval would measure class assignment rather than sample identification.

InfoNCE loss. For a batch of N matched image–text pairs $\{(I_i, T_i)\}_{i=1}^N$, the model embeds each side into the shared d -dimensional space and produces two L2-normalised batches of embeddings, $\{\mathbf{v}_i\}$ and $\{\mathbf{t}_i\}$. Define the scaled cosine similarity between image i and prompt j as

$$s_{ij} = \frac{\mathbf{v}_i \cdot \mathbf{t}_j}{\tau}, \quad (3.1)$$

where τ is the learnable temperature introduced below. For each image i the image-to-text contrastive loss treats the matched prompt i as the positive and the other $N-1$ prompts in the same batch as negatives; the row-wise softmax cross-entropy then reads

$$\mathcal{L}_{I \rightarrow T} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(s_{ii})}{\sum_{j=1}^N \exp(s_{ij})}. \quad (3.2)$$

The symmetric InfoNCE loss [?, 2] averages this with the analogous column-wise text-to-image loss $\mathcal{L}_{T \rightarrow I}$ (in which each prompt is matched against its image positive and the $N-1$ batch images as negatives):

$$\mathcal{L}_{\text{InfoNCE}} = \frac{1}{2} (\mathcal{L}_{I \rightarrow T} + \mathcal{L}_{T \rightarrow I}). \quad (3.3)$$

Geometrically, minimising $\mathcal{L}_{\text{InfoNCE}}$ pulls matched (image, prompt) pairs together in cosine-similarity space and pushes the $N-1$ batch-internal mismatches apart.

Learnable temperature. The scalar τ in equation (3.1) controls the sharpness of the softmax in equation (3.2): a smaller τ peaks the distribution and pushes the model toward sharply confident positives, while a larger τ smooths it. Following CLIP [2] we store $\log(1/\tau)$ as the trainable variable (which guarantees $\tau > 0$ throughout training without an explicit constraint) and initialise it to $\log(1/0.07) \approx 2.66$, the CLIP-paper default. Gradient descent moves it freely during training. Empirically $\log(1/\tau)$ drifts upward (i.e. τ shrinks) as the embedding space stabilises, reflecting the model’s preference for sharper positives as the contrastive task becomes easier; the converged value is reported alongside the loss curves in Chapter 4.

Caveats. The InfoNCE objective above assumes the $N-1$ in-batch negatives are *genuine* mismatches. With our 5-class LC25000 supervision and batch size $N=32$ (Table 3.2), the expected number of true mismatches per image is $0.8 \times (N-1) \approx 24.8$ but the expected number of *false negatives* (other batch entries sharing the image’s class, and therefore its prompt) is ~ 6 . The loss therefore penalises some semantically-correct alignments. We do not mitigate this in the present study — it is documented as a limitation in Chapter 5.

Figure 3.1 reproduces the canonical CLIP paradigm as introduced by Radford et al. [2]: contrastive image–text pretraining (panel 1) builds a shared embedding space; class prompts then become the zero-shot classifier (panels 2–3). Figure 3.2 is the specific instantiation of that paradigm used in this thesis — short histopathology class prompts, 256-d projection heads on top of frozen foundation-model encoders, and a single fixed prompt strategy used at both training and evaluation time.

3.3.3 PLIP: a histopathology-pretrained CLIP-style model

- Huang et al. [7] (2023) — a CLIP-style model pretrained on histopathology image–text pairs.
- Training corpus: OpenPath, $\sim 208\text{k}$ histopathology image–text pairs scraped from Twitter conversations among pathologists, supplemented with LAION subset filtering.
- Architecture: CLIP ViT-B/32 vision encoder paired with a CLIP-style text encoder, both trained contrastively from a CLIP initialisation.
- Used in this thesis in two distinct ways:
 - As a *zero-shot reference baseline* (§3.6): the full PLIP model (image + text encoders) used off-the-shelf for cosine-similarity classification on all four datasets (LC25000 + the three external evaluation sets).
 - As a *pretrained image encoder* in our image-encoder ablation (§3.4.2): PLIP’s ViT-B/32 vision encoder paired with our trained projection head and a separate BERT text encoder.

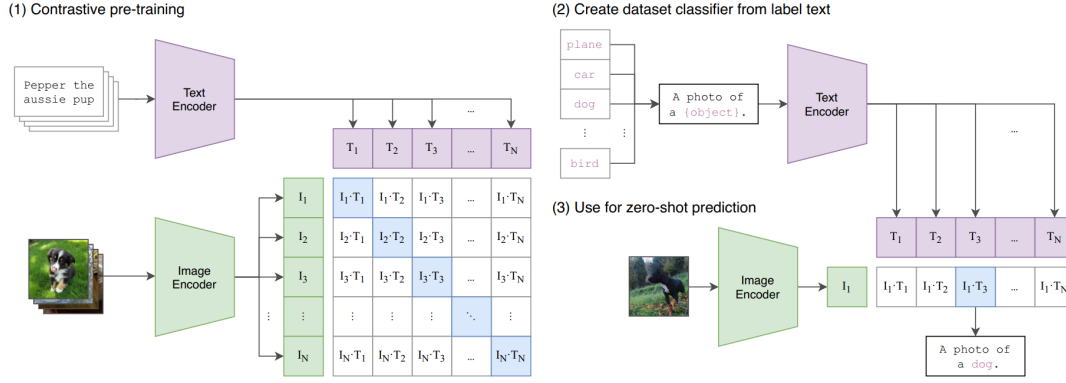


Figure 3.1 The CLIP paradigm as introduced by Radford et al., 2021 (Figure 1 of the original paper, reproduced for illustration). **(1) Contrastive pre-training:** two encoders project a batch of image–text pairs into a shared embedding space; the in-batch cosine-similarity matrix is supervised so the diagonal pairs (matched) score higher than off-diagonals (mismatched). **(2) Build a dataset-specific classifier from label text:** any new task is cast as text by encoding class-name prompts through the same text encoder. **(3) Zero-shot prediction:** an unseen image is classified by cosine-similarity argmax against the class prompts — no task-specific training needed.

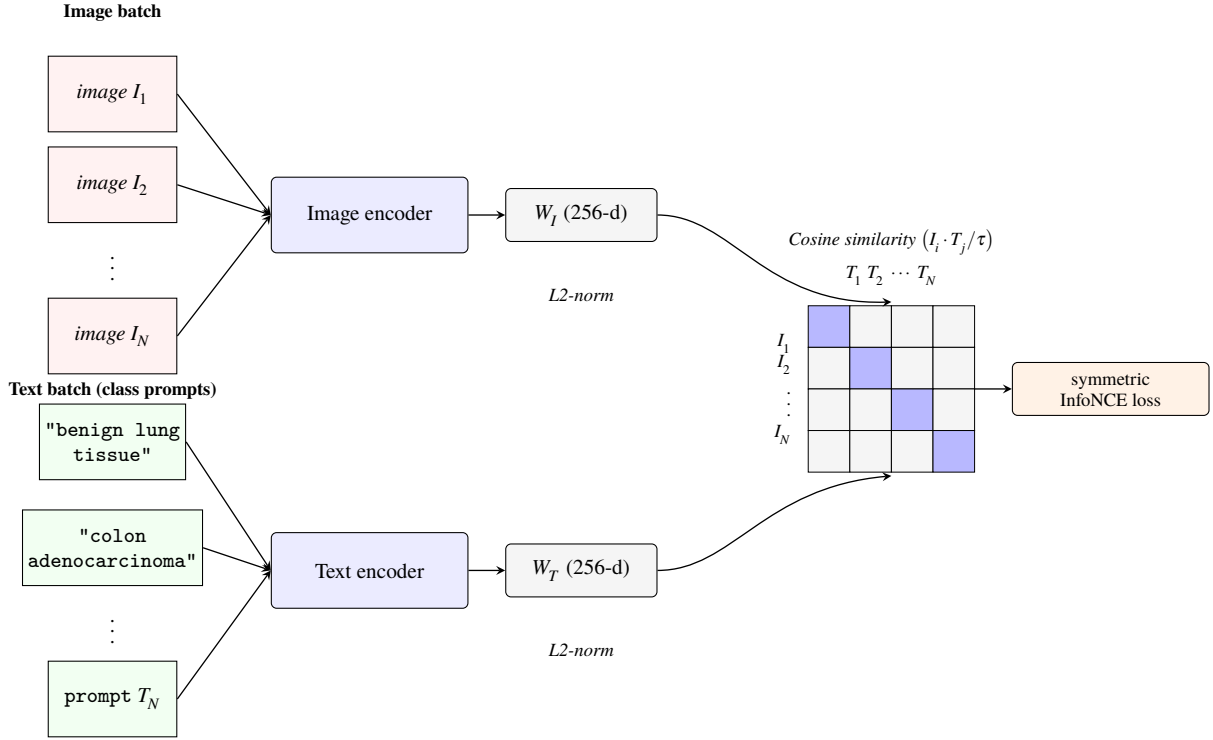


Figure 3.2 CLIP dual-encoder architecture used in this project. Two independent encoders project a batch of N image–text pairs into a shared d -dimensional embedding space (here $d = 256$, the projection head’s output). After L2-normalisation and a learnable temperature τ , the $N \times N$ cosine-similarity matrix between every image and every prompt is the input to the symmetric InfoNCE loss. The diagonal (blue) entries are the positive pairs; off-diagonal entries are negatives within the batch.

3.4 Image Encoders

3.4.1 ResNet50 (CNN, ImageNet supervised)

- He et al. [11] (2016), residual learning.
- $\sim 23\text{M}$ parameters, 2048-d output via global average pooling.
- Supervised pretraining on ImageNet-1K ($\sim 1.3\text{M}$ natural images, 1000 classes).
- Role in this thesis: general-purpose image-encoder baseline; carries the assumption that ImageNet features transfer broadly.

3.4.2 Vision Transformer (ViT)

- Dosovitskiy et al. [12] (2020).
- Patch-based architecture: image split into $N \times N$ patches, each linearly embedded as a token, then processed by a stack of transformer encoder layers with self-attention.
- The “B” sizing (ViT-Base): 12 layers, hidden size 768.
- The “/16”, “/32”, “/14” suffixes denote patch size; smaller patch = more tokens = finer spatial granularity at the same input resolution.
- Role in this thesis: we use **PLIP’s pretrained ViT-B/32** as one image-encoder variant. PLIP itself is described in §3.3.3; this section covers only the ViT architecture used by PLIP’s vision encoder.

3.4.3 DINOv2 (self-supervised on general natural images)

- Oquab et al. [13] (2024).
- Self-supervised pretraining on LVD-142M ($\sim 142\text{M}$ curated general natural images); no labels, no text supervision, no histopathology data.
- Self-distillation between a teacher and student network on augmented crops of the same image.
- We use the base size with patch 14 (**ViT-B/14**).
- Role in this thesis: a control. Same general ViT architecture family as PLIP’s vision encoder, similar self-supervised paradigm, but no histopathology pretraining — isolates “ViT + self-supervision” from “histopathology pretraining specifically” in the image-encoder ablation.

3.5 Text Encoders

- Transformer [27] + BERT [10] — text encoder used by baseline.
- BioBERT [8] / PubMedBERT [9] — biomedical text encoder swap ablation.

3.6 Reference baselines: PLIP zero-shot and a plain image classifier

3.6.1 PLIP zero-shot

- Uses the full PLIP model (§3.3.3) — both image and text encoders, frozen — for zero-shot classification.
- Class prediction: cosine similarity between the image embedding and the five PLIP-encoded LC25000 class-prompt embeddings, argmax.
- No LC25000 supervision; no projection head trained on our data.

- Evaluated on all four datasets: LC25000 (training source) plus the three external evaluation datasets (NCT-CRC, Chaoyang, LungHist700; the latter two are queued follow-up runs at the time of writing).
- Role: isolates the value of histopathology pretraining alone, without any of our fine-tuning.

3.6.2 Plain image classifier

- Drops the CLIP paradigm entirely: no text encoder, no projection heads, no learnable temperature, no contrastive loss.
- Architecture: ResNet50 → Global Average Pooling → BatchNorm → Dense(5) trained with sparse cross-entropy.
- Role: isolates what the CLIP paradigm itself contributes once the image encoder, data and adaptation budget are held constant.

3.7 Training Dataset: LC25000

- Borkowski et al. [3] (2019).
- 5 tissue classes (lung, colon).
- 25 000 patches from 1 250 originals via rotation/flip → no further geometric augmentation.
- Known limits: single institution, possible stain artifacts, no patient-level grouping.

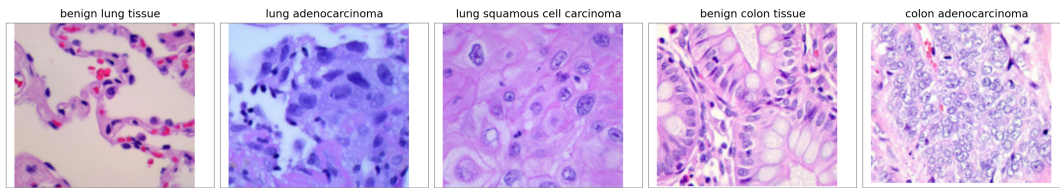


Figure 3.3 Example image per class from LC25000. Left-to-right: benign lung tissue, lung adenocarcinoma, lung squamous cell carcinoma, benign colon tissue, colon adenocarcinoma. The two benign classes appear visually distinct from the cancer classes; within the cancers, lung adenocarcinoma vs lung squamous cell carcinoma is the hardest discrimination in our experiments (and a clinical confound).

Table 3.1 LC25000 class counts and stratified split sizes.

Class	Total	Train	Val	Test
benign lung tissue	5 000	4 000	500	500
lung adenocarcinoma	5 000	4 000	500	500
lung squamous cell carcinoma	5 000	4 000	500	500
benign colon tissue	5 000	4 000	500	500
colon adenocarcinoma	5 000	4 000	500	500
Total	25 000	20 000	2 500	2 500

3.8 External Evaluation Datasets

- Role: data the model has never seen during training. We evaluate every LC25000-trained variant against three external datasets, chosen to span shift severity and two organs.
- Order presented below corresponds to increasing shift severity relative to LC25000.

3.8.1 NCT-CRC-HE-7K

- Kather et al. [4] (2019), Zenodo record 1214456.
- 7 180 patches across 9 tissue classes; we restrict to NORM ($\leftrightarrow$ benign colon tissue) and TUM ($\leftrightarrow$ colon adenocarcinoma) for 1 974 patches scored (741 NORM + 1 233 TUM).
- 224×224 patches at $\sim 20\times$ magnification, Macenko-normalised at release.
- Role: close-shift colon evaluation (stylistically similar acquisition to LC25000).

3.8.2 Chaoyang

- Zhu et al. [5] (2022; preprint 2021).
- 6 160 patches across 4 classes (normal, serrated, adenocarcinoma, adenoma); we restrict to normal ($\leftrightarrow$ benign colon, $n = 1 816$) and adenocarcinoma ($\leftrightarrow$ colon adeno, $n = 2 244$) for 4 060 patches scored.
- 512×512 patches at $20\times$, Beijing Chaoyang Hospital.
- Documented label noise in the training split; the test split is labelled by consensus of three pathologists.
- Role: different-hospital colon evaluation (different scanner, different staining, label noise).

3.8.3 LungHist700

- Tabatabaei et al. [6] (2024, *Scientific Data*).
- ~ 691 high-resolution images at $20\times$ and $40\times$ covering normal lung tissue plus adenocarcinoma and squamous-cell carcinoma at three differentiation grades each; we restrict to the $20\times$ subset for 3-class scoring (359 images: 85 normal + 146 AdC + 128 SqC).
- Role: different-source lung evaluation; the only multi-class scoring among the external datasets.

3.9 Preprocessing and Input Pipeline

3.9.1 Image preprocessing (per-backbone)

- ResNet50: RGB / 255.0 (i.e., $[0, 1]$ linear scaling).
- ViT-B/32 (PLIP): CLIP-standard mean/std normalisation
mean = $[0.481, 0.458, 0.408]$, std = $[0.269, 0.261, 0.276]$.
- ViT-B/14 (DINOv2): ImageNet mean/std normalisation
mean = $[0.485, 0.456, 0.406]$, std = $[0.229, 0.224, 0.225]$.
- All inputs resized to 224×224 bilinear. No additional augmentation (LC25000 already includes rotation/flip augmentation at release).
- Each backbone expects the input distribution it was pretrained on; mismatched preprocessing produces degraded features — see §3.9.4.

3.9.2 Text preprocessing

- All text encoders use BERT-family WordPiece tokenizers (BERT, BioBERT, PubMedBERT, and PLIP's text encoder).
- Fixed sequence length 24 tokens (covers all class-prompt formats).
- Composed prompt format: "A histopathology image of {class_name}."

3.9.3 LC25000-specific cleanup

- 1 280 byte-identical duplicate files identified by MD5 grouping; excluded via `src/data/lc25000_dedupe_exclusions.json`.
- Stratified 80/10/10 split, seed 42, persisted to canonical JSON.
- The canonical split file predates the dedupe pass, so the 1 280 duplicates still cross train/test for the runs reported; flagged as a limitation in Chapter 5.

3.9.4 Methodological note on preprocessing

- Mid-project we caught that the pipeline was applying Caffe-style BGR preprocessing to a backbone expecting $[0, 1]$ RGB — a mismatch from two libraries shipping different ResNet50 checkpoints under the same name.
- The error was surfaced via a linear-probe diagnostic on the frozen features (the probe macro-F1 was well above the trained CLIP ceiling), and fixed with a centralised image loader at `src/data/images.py`.
- Beyond the project-specific bug, the takeaway is that linear-probing a frozen feature extractor against the trained model’s ceiling is a cheap sanity check worth running as a project-foundation step.

3.10 The Baseline

- Frozen ResNet50 + frozen BERT.
- Projection heads (Dense + LayerNorm) $\rightarrow$ 256-d shared space.
- Learnable temperature, CLIP-style init.
- Symmetric InfoNCE, AdamW, grad clipping.
- Single fixed prompt at train + eval.
- Float32 throughout.

Table 3.2 Baseline hyperparameters. Ablations change exactly one row.

Hyperparameter	Value
Image encoder	ResNet50 (ImageNet preset), frozen
Text encoder	BERT base uncased, frozen
Projection head	Dense(256, no bias) $\rightarrow$ LayerNorm
Embedding dimension	256
Image size	224×224
Text sequence length	24 WordPiece tokens
Batch size	32
Initial temperature	$\tau = 0.07$
Loss	symmetric InfoNCE
Optimizer	AdamW
Learning rate	1×10^{-3}
Weight decay	1×10^{-4}
Gradient clipnorm	1.0
Max epochs	30
EarlyStopping patience	5
ReduceLROnPlateau	factor 0.5, patience 2, min 10^{-6}
Numerical precision	float32
Random seed	42

3.11 Evaluation Protocol

3.11.1 Metrics and visualizations

- Classification rule (CLIP variants): argmax of cosine similarity between the image embedding and the five LC25000 class-prompt embeddings.
- Classification rule (plain classifier): argmax of softmax over Dense(5).
- Global metrics: accuracy, balanced accuracy, macro-F1, weighted F1.
- Per-class: precision, recall, F1, support.
- Confusion matrices: raw counts + row-normalised by true class.
- Visualizations: shared image + prompt UMAP (training source), and cross-distribution UMAP grid per external dataset (LC25000 dots + external triangles overlaid in one fit).

3.11.2 External-dataset evaluation: restricted-argmax protocol

- Each variant exposes all 5 LC25000 class prompts at inference, but on external evaluation we argmax *only* over the LC25000 classes that map to the external dataset's organ:
 - NCT and Chaoyang (colon datasets): argmax restricted to {benign colon tissue, colon adenocarcinoma}.
 - LungHist700 (lung dataset): argmax restricted to {benign lung tissue, lung adenocarcinoma, lung squamous cell carcinoma}.
- Why: external datasets cover only a subset of LC25000 classes; an unrestricted argmax would routinely predict the model's strongest out-of-organ class on these images and produce uninformative misclassifications.
- Predictions into a non-target LC25000 class count as wrong but are reported separately when informative for confusion analysis.
- Pure inference: every variant reuses its LC25000-trained checkpoint; no fine-tuning on external sets.
- Class-mapping rules per dataset are detailed in §3.8.

3.11.3 Per-organ reporting

- Results are reported per external dataset, not averaged across datasets.
- Aggregating across datasets would hide the cross-organ regime change observed in this study (Chapter 4); a single mean-OOD number collapses an order-of-magnitude difference between organs into one misleading point.
- Per-dataset reporting is therefore an evaluation choice with explicit methodological motivation.

3.12 Experimental Design and Ablation Protocol

- Single-variable-change principle.
- Matrix grouped by axis of variation; full matrix in Table 4.1 of Chapter 4.
- Scope boundaries: no patient-level grouping, no human-expert evaluation, no architecture-controlled comparison vs CONCH [20] / BiomedCLIP [22] / MUSK [23]. Deferred to future work.

3.13 Computational Setup and Reproducibility

- Hardware: Google Colab T4 (occasional L4 for longer runs); local CPU/GPU fallback for development and analysis.
- Main stack: TensorFlow + KerasHub for the CLIP-style training pipeline and all downstream evaluation.
- Auxiliary stack: PyTorch + HuggingFace transformers for image-encoder feature extraction when the encoder lacks a TensorFlow port (DINOv2; see below).
- Library versions pinned in `requirements.txt`.
- Reproducibility: seed 42 across NumPy, TensorFlow, and PyTorch; `PYTHONHASHSEED` set; `tf.config.experimental.enable_op_determinism()`. Runs are bit-exact on the same hardware; small numerical drift can appear across GPU types but does not change the classification outcomes reported.
- DINOv2 has no TensorFlow port in our environment; its image features are precomputed once in PyTorch and cached to disk as HDF5 (one file per dataset, keyed by image relative path). The TensorFlow training and evaluation pipelines then read from cache. Mathematically equivalent to running the frozen backbone online, since the backbone is frozen and each image always produces the same feature vector. Total cache footprint ~ 100 MB across LC25000 plus the three external datasets.

4 Results and Discussion

4.1 Experimental Overview

4.1.1 Research questions

The chapter is built around six research questions, each targeting a single design axis of a CLIP-style histopathology pipeline: the baseline reference, the CLIP-versus-non-CLIP choice, the input-side preprocessing, the text encoder, the image encoder, and cross-dataset generalisation. Five of them (RQ1–RQ5) are answered by a dedicated experiment section in this chapter; RQ6 is cross-cutting and is addressed by the external evaluations run alongside every other experiment. The experiment matrix in Table 4.1 below maps each RQ to the controlled comparison that addresses it.

- **RQ1** — What baseline classification and generalisation behaviour does a frozen general-purpose CLIP pipeline achieve on LC25000, and what failure modes appear?
- **RQ2** — Does the CLIP paradigm validate its *foundational-model* premise — learning one task and generalising to tasks it was not specifically trained for — relative to a non-CLIP classifier on the same image encoder?
- **RQ3** — Does stain normalisation, an input-side preprocessing that targets a known shortcut in histopathology, help when evaluating on datasets the model was never trained on?
- **RQ4** — Does using a text encoder pretrained on biomedical text (BioBERT, PubMedBERT), instead of generic BERT, improve performance under short class-name prompts?
- **RQ5** — Does the image encoder’s *pretraining choice* matter? We test two alternatives to the ImageNet-supervised ResNet50 baseline: (a) a Visual Transformer pretrained on histopathology image–text pairs (PLIP), and (b) a Visual Transformer pretrained self-supervised on general natural images (DINOv2), which serves as a control to separate the contribution of “ViT architecture and self-supervision” from “histopathology pretraining specifically”.
- **RQ6** — How well does each variant generalise to data the model has never seen? We evaluate every variant against three external datasets across two organs that span shift severity: close-shift colon, different-hospital colon, and different-source lung.

4.1.2 Training source and external datasets

Every variant in this chapter is trained on a single source dataset and evaluated on three external datasets it has never seen during training. The training source is LC25000 [3], already introduced in §4.2.1: every trained variant is fit on it, and the in-distribution test split yields the macro-F1 we refer to as *training-source results* throughout the chapter. The three external datasets — the source of every *generalisation* number in the chapter — are listed in order of shift severity:

- **NCT-CRC-HE-7K** [4], close-shift colon: 1 974 scored patches over the NORM and TUM classes, mapping to *benign colon tissue* and *colon adenocarcinoma* in LC25000.

Table 4.1 Experiment matrix. Each experiment isolates exactly one axis of variation relative to the baseline. Sections §4.2–§4.7 follow this order, ending with image-encoder pretraining (the chapter’s headline finding).

Experiment	Image encoder	Text encoder	Loss	Preprocessing	Axis tested
Baseline (§4.2)	ResNet50 (ImageNet)	BERT base	InfoNCE	RGB / 255	reference
Stain normalization (§4.3)	ResNet50 (ImageNet)	BERT base	InfoNCE	Macenko / Reinhard / Vahadane	input-side
PLIP zero-shot (§4.4)	ViT-B/32 (PLIP / OpenPath)	PLIP text encoder	none (zero-shot)	CLIP mean/std	reference: no LC25000 supervision
Foundational vs plain (§4.5)	ResNet50 (ImageNet)	— (no text)	cross-entropy	RGB / 255	architecture: CLIP paradigm vs plain classifier
Text encoder (§4.6)	ResNet50 (ImageNet)	BioBERT / PubMedBERT	InfoNCE	RGB / 255	text encoder pretraining
Image encoder (§4.7)	ViT-B/32 (PLIP); ViT-B/14 (DINOv2)	BERT base	InfoNCE	varies	image encoder pretraining

- **Chaoyang** [5], different-hospital colon: 4060 scored patches (normal and adenocarcinoma; the serrated and adenoma classes have no clean LC25000 analogue and are excluded from scoring), mapping to the same two LC25000 colon classes as NCT.
- **LungHist700** [6], different-source lung: 359 images at $20\times$ magnification (the $40\times$ subset is excluded), mapping to LC25000’s *benign lung tissue*, *lung adenocarcinoma*, and *lung squamous cell carcinoma*.

Full dataset provenance, acquisition details, and per-dataset preprocessing live in Methodology (§3.8); the bullets above keep only what is needed to interpret the external-evaluation tables in the rest of the chapter.

Three protocol choices apply to every external evaluation that follows. First, **restricted argmax**: each variant exposes all 5 LC25000 class prompts at inference, but the argmax is restricted to the LC25000 classes that map to the external dataset’s organ (2 colon classes for NCT and Chaoyang, 3 lung classes for LungHist700). Predictions into a non-target LC25000 class count as wrong but are reported separately when informative for confusion analysis. Second, **pure inference**: every variant reuses its LC25000-trained checkpoint without further adaptation; no fine-tuning is performed on the external datasets. Third, **per-dataset reporting**: results are presented per dataset rather than averaged. The motivation for that last choice is methodological — aggregating across datasets would average a colon-side partial transfer (~ 0.79 – 0.87 F1) with a lung-side categorical failure (0.305 F1 in the baseline), producing a single mean-OOD number that hides the cross-organ regime change that ends up driving the chapter’s headline result in §4.7.

4.1.3 Metrics and visualizations

Each variant in this chapter is reported with the same set of metrics and visualisations. The choices and what each is meant to surface:

- **Macro-F1** is the headline metric. It averages the per-class F1 scores equally, which is the right behaviour for the per-class precision–recall trade-offs that drive the chapter’s substantive comparisons. Formal definitions are in Methodology (§3.11).
- **Accuracy, balanced accuracy, weighted F1, and per-class precision/recall/F1** are reported alongside macro-F1 for completeness, but the chapter’s interpretive weight is on macro-F1 and the per-class numbers.
- **Confusion matrices** (raw counts and row-normalised by true class) are reported per variant and per dataset — the most direct view of how a model distributes its errors across the available classes.
- **Training-source UMAP** projects LC25000 image embeddings and class-prompt embeddings into the same 2-D fit (cosine metric), letting the reader see cluster separation and the alignment between image clusters and prompt anchors.
- **Cross-distribution UMAP grid** (per external dataset, per variant) overlays LC25000 dots and external-dataset triangles in a single 2-D fit, asking whether the external dataset co-locates with the matching LC25000 class cluster or sits in its own region of feature space.

4.2 Baseline: frozen ResNet50 and frozen BERT

4.2.1 Configuration

The baseline pipeline pairs a frozen ResNet50 image encoder, taken from the standard ImageNet-supervised checkpoint, with a frozen BERT-base-uncased text encoder; only a pair of small projection heads — a single 256-wide Dense layer with no bias, followed by LayerNorm — and a scalar temperature τ are trained on top. This is deliberately the most general-purpose CLIP one can assemble. ResNet50 is the canonical CNN baseline of the past decade and BERT-base is the canonical text encoder; neither has been exposed to histology during pretraining, and neither has been tuned for the task. The reason to start here is mechanical: every subsequent variant in this chapter changes exactly one piece of this configuration, and the resulting delta is only interpretable if the reference is something recognisable rather than something already adapted to the data.

Everything else is held constant across the variants that follow. The training data is the LC25000 dataset, with a 80/10/10 train-validation-test split at random seed 42; a single fixed composed prompt, “A histopathology image of {class_name}.”, is used at both training and inference; the loss is symmetric InfoNCE; the optimiser is AdamW with learning rate 1×10^{-3} and weight decay 1×10^{-4} ; and training runs for at most 30 epochs, with EarlyStopping (patience 5, restore best weights) terminating it earlier on every run. The remaining design choices — batch size, image and sequence dimensions, initial temperature, gradient clipping, learning-rate schedule, numerical precision, weight initialisation — are not referenced again in this chapter and are gathered in the baseline hyperparameter table in Methodology (Table 3.2).

4.2.2 Training behaviour and in-distribution results

On the LC25000 test split, the baseline reaches a macro-F1 of 0.996. The model has only two trainable projection heads and a scalar temperature τ on top of frozen encoders — around 720,000 parameters in total — so a saturated in-distribution score would invite a memorisation reading without a check on the training dynamics. Figure 4.1 provides that check: the training and validation InfoNCE losses (equation (3.3)) descend together over the 30-epoch budget and stay close at convergence. The learnable temperature τ (equation (3.1)), initialised at the CLIP-paper value of 0.07, *rises* during training, peaks at ≈ 0.090 around epoch 10, and settles near 0.087. This is the opposite direction from the CLIP paper’s $\tau \rightarrow 0.01$ behaviour on uniquely-paired image-text data, and is consistent with our shared-prompt setup: when multiple images per batch share the same class prompt (the false-negative caveat in §3.3), softening τ reduces the penalty those false negatives impose on otherwise-correct alignments.

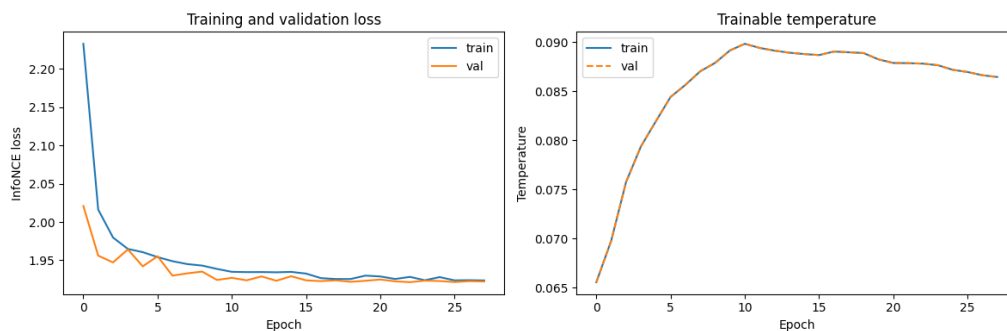


Figure 4.1 Baseline training (blue) and validation (orange) InfoNCE loss across epochs (left), and the learnable temperature τ on the same horizontal axis (right).

The embedding space the model converges to is shown in Figure 4.2. Each LC25000 class forms a distinct cluster of image embeddings and the five class-prompt anchors (encoded through frozen BERT and projected to the same 256-d space) sit at the centre of their corresponding clusters — the expected post-InfoNCE alignment between text and image, with the trained projection heads doing the work that the frozen text encoder alone could not. The lung-adenocarcinoma cluster sits closest to lung squamous cell carcinoma in the 2D projection, foreshadowing the off-diagonal confusion-matrix mass discussed below.

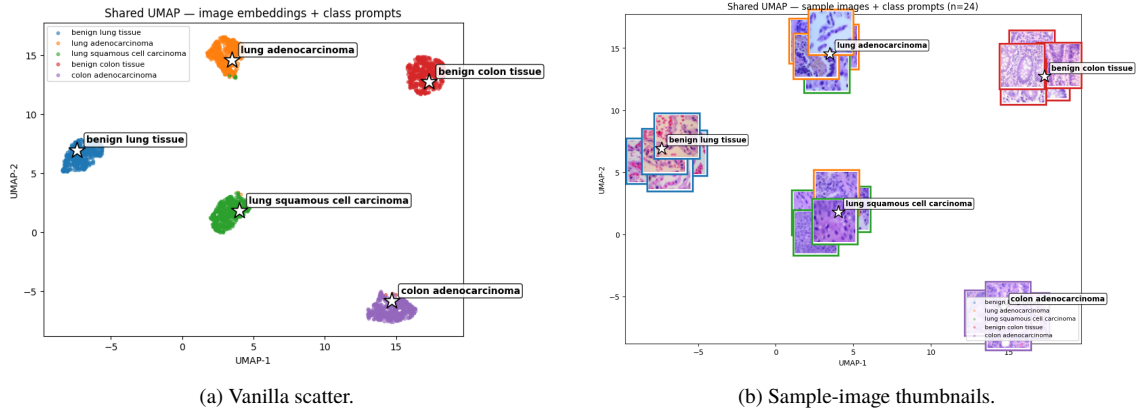


Figure 4.2 UMAP of the baseline embedding space on the LC25000 test split. Two views of the same fit: (a) per-class scatter with the five class-prompt anchors marked, (b) the same fit overlaid with sample-image thumbnails.

The confusion matrix (Figure 4.3) confirms that the saturated 0.996 is genuinely close to ceiling: the benign tissues and lung squamous cell carcinoma separate cleanly, with the off-diagonal mass concentrated on lung adenocarcinoma — the hardest discrimination in the LC25000 schema. In-distribution, thus, does not discriminate the baseline from any of the trained variants studied in subsequent sections (all of them saturate at ~ 0.99); the meaningful comparison is the external evaluation that follows in §4.2.3.

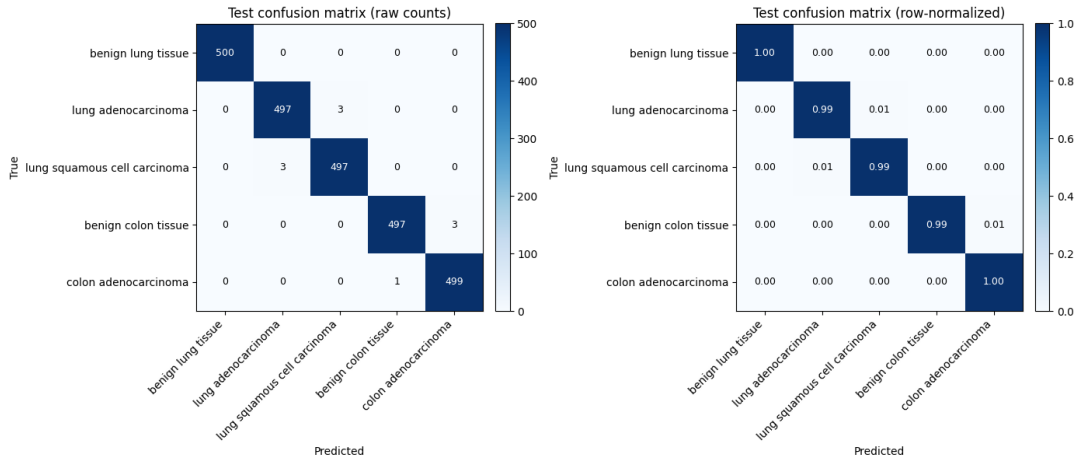


Figure 4.3 Baseline confusion matrix on the LC25000 test split. Left: raw counts. Right: row-normalised by true class.

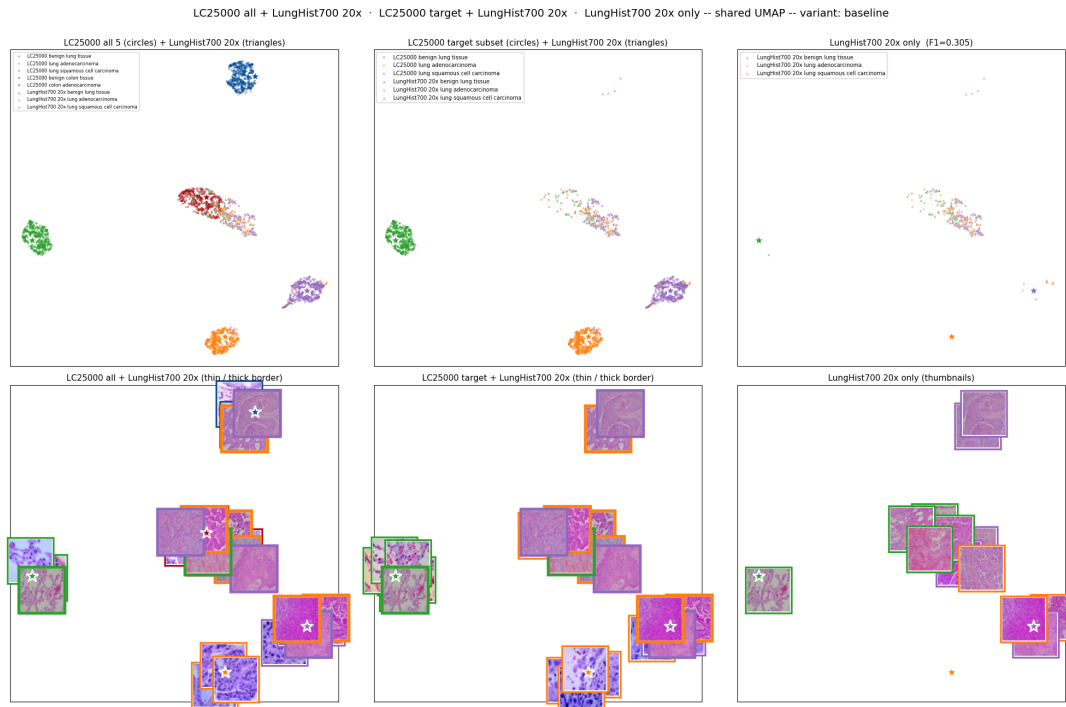
4.2.3 Generalization to external datasets

The baseline’s macro-F1 on each external dataset, restricted-argmax over the target-organ LC25000 classes, is reported in Table 4.2. The result splits in two. On the two colon datasets the baseline transfers partially — 0.866 on NCT-CRC-HE-7K, the dataset stylistically closest to LC25000’s colon distribution, and 0.786 on Chaoyang, where a different hospital, scanner, and staining protocol drag the score down by ~ 0.08 F1 even though the organ is unchanged. On lung the picture is qualitatively different: **0.305 macro-F1 on LungHist700’s 3-class problem, where chance is 0.33**. The colon-versus-lung gap dwarfs every within-organ shift in the table — the central observation the rest of the chapter sets out to explain.

Table 4.2 Baseline generalization to the three external datasets, restricted-argmax over the target-organ LC25000 classes.

External dataset (organ)	Macro-F1	Accuracy	Bal. acc.	Weighted F1
NCT-CRC-HE-7K (colon)	0.866	0.870	0.866	0.871
Chaoyang (colon)	0.786	0.796	0.782	0.791
LungHist700 (lung, 20×)	0.305	0.390	0.383	0.301

The lung results are not a noisy degradation of the colon transfer; they are a different failure mode. Per-class recall on LungHist700 breaks down as 0.89 for lung squamous cell carcinoma, 0.07 for lung adenocarcinoma, and 0.19 for benign lung tissue. Back-calculated from the precision figures, **the model assigns “squamous” to roughly 84% of LungHist700 inputs regardless of true class** — and given that squamous accounts for only 36% of the ground truth, the near-chance macro-F1 is the direct consequence of that bias. Figure 4.4 shows the matching feature-space signature: all 359 LungHist700 samples collapse into a single tight cluster with no visible internal class structure. The frozen ImageNet-supervised ResNet50, never exposed to histology during pretraining, extracts no usable per-class variance from LungHist700 tissue, and the trained projection head has nothing to align with the three lung class prompts.

**Figure 4.4** Baseline cross-distribution UMAP on LungHist700. LC25000 image embeddings (dots) and LungHist700 image embeddings (triangles) projected into the same 2-D fit.

The Chaoyang dataset sits between these two regimes (Figure 4.5). The two colon classes are partially separated in the projected space, and the Chaoyang triangles cluster in a region adjacent to — but not co-located with — the matching LC25000 colon clusters: enough signal for partial transfer, with a measurable cost from the cross-hospital distribution gap. Together with NCT and LungHist700, this exposes two distinct generalisation regimes for the experiments that follow. There is a colon regime, where frozen ImageNet features carry good-enough structure for partial transfer, and a lung regime, where they carry no usable structure at all. The central question the rest of this chapter is built around: is the lung-side failure about the encoder’s architecture, about its supervised pretraining on natural images, or about the absence of histopathology data in pretraining? The DINOv2 control (general-domain self-supervised pretraining, no histology) and the PLIP backbone (ViT pretrained on histopathology image–text pairs) jointly decompose that question, and the headline answer arrives in §4.7.

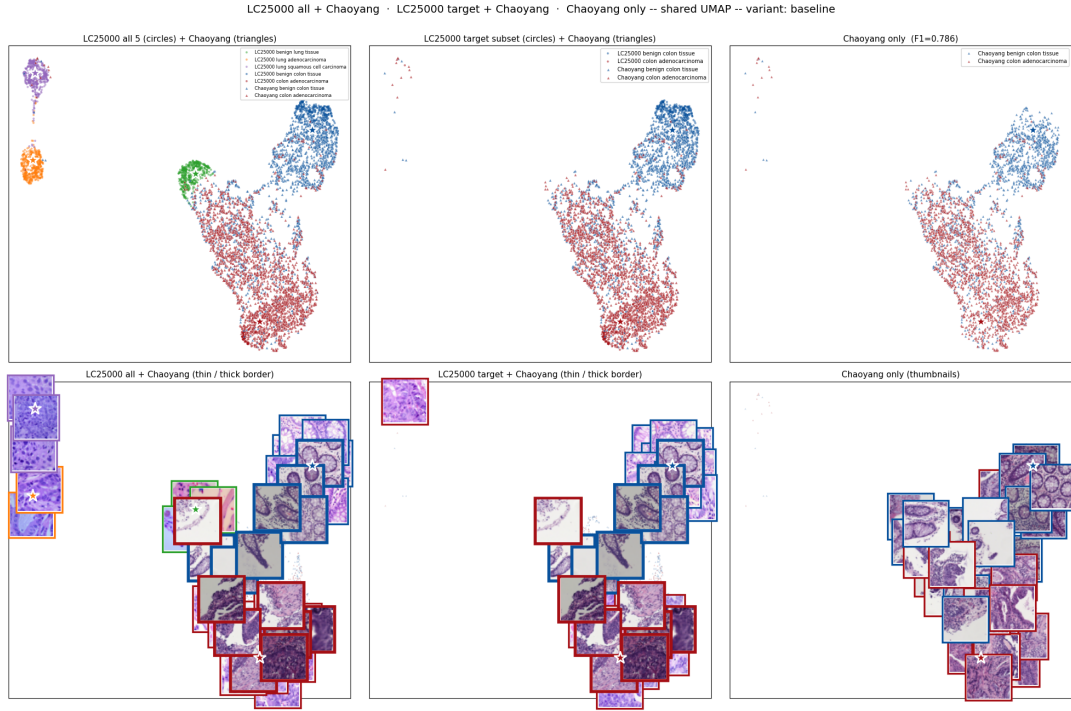


Figure 4.5 Baseline cross-distribution UMAP on Chaoyang. LC25000 image embeddings (dots) and Chaoyang image embeddings (triangles) projected into the same 2-D fit.

4.3 Stain normalization

4.3.1 Configuration

The stain-normalisation experiment keeps the baseline pipeline architecture identical to §4.2.1 and changes only the image-input distribution. Every LC25000 training image is Macenko-normalised [24] against a single fixed reference patch: the first NORM tile from NCT-CRC-HE-7K, abbreviated “NCT-NORM” below.

The choice of reference matters operationally. NCT-CRC is distributed already Macenko-normalised against an unpublished Kather-team reference [4]; by Macenko-normalising LC25000 onto an NCT-CRC tile, we land the training distribution in approximately the colour space NCT-CRC arrives in at release, which lets NCT-CRC be fed raw at evaluation and avoids the double-Macenko regime that historically zeroed out the OOD F1 in an earlier version of this experiment. Chaoyang and LungHist700, by contrast, arrive in their native un-normalised palettes, so for those two datasets the same Macenko(NCT-NORM) transform is applied at inference to bring them into the model’s training-stain space. An alternative reference drawn from Chaoyang or LungHist700 — the latter being the most stain-distinct of the three external datasets — would have re-introduced the double-Macenko problem on NCT-CRC and required a separate retraining run for each reference tested; the NCT-NORM reference is therefore picked for methodological consistency rather than per-dataset optimisation, and reference rotation as an ablation axis is documented as a limitation in Chapter 5.

Reinhard and Vahadane were also explored as alternative normalisations in an exploratory pass but did not advance to the full external-evaluation ablation. Vahadane’s NMF solver fails to converge on a fraction of LC25000 patches — visible in Figure 4.6(d), where the bottom-row colon_aca cell re-renders near-blank. Reinhard is a Lab-space colour-statistics transfer rather than a stain-decomposition method, included in the figure for visual comparison only.

4.3.2 Generalization to external datasets

On LC25000 the Macenko-trained checkpoint saturates at 0.986 macro-F1, alongside the baseline and every other trained variant in this study — in-distribution does not discriminate this lever, so the substantive

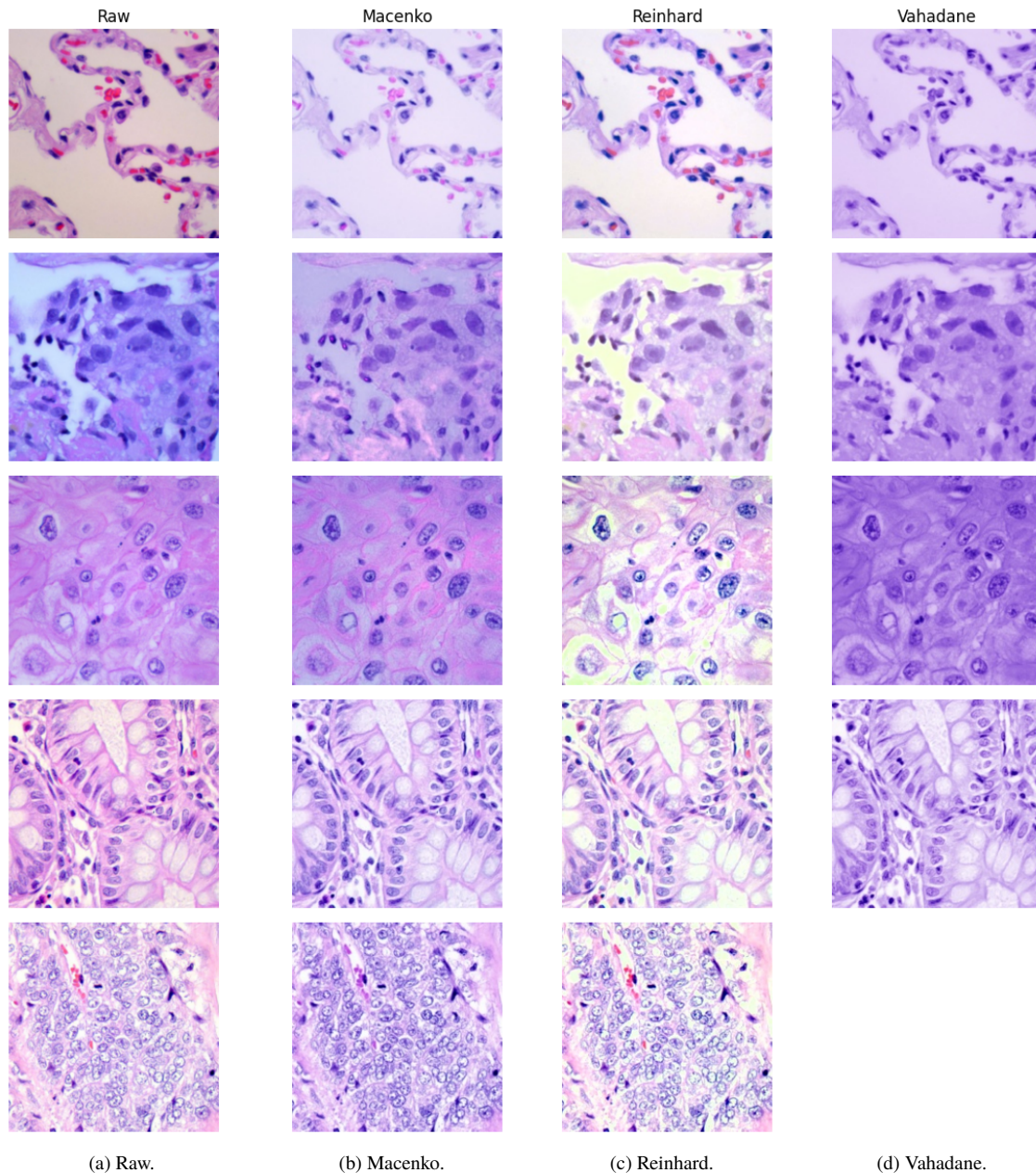


Figure 4.6 Stain normalization, one example per class. Column (a) shows the raw inputs (identical across methods, shown once for compactness); columns (b)–(d) show the same inputs after Macenko, Reinhard, and Vahadane normalization respectively. The bottom-row colon_aca cell in column (d) shows the Vahadane NMF solver failing to converge and re-rendering near-blank, an instability discussed in the interpretation.

comparison is the external evaluation reported in Table 4.3.

Table 4.3 Macenko(NCT-NORM) vs baseline on the three external evaluation datasets.

External dataset	Baseline F1	Macenko F1	Delta
NCT-CRC-HE-7K (close-shift colon)	0.866	0.844	−0.022
Chaoyang (different-hospital colon)	0.786	0.816	+0.030
LungHist700 (different-source lung, 20×)	0.305	0.325	+0.020

The lever’s direction is data-dependent. On the two datasets that arrive in their native stain palette —

Chaoyang (+0.030 F1) and LungHist700 (+0.020 F1), both receiving Macenko at inference — normalisation moves the input closer to the model’s training distribution and the frozen ImageNet ResNet50 produces better features. On NCT-CRC the picture inverts: NCT was already in approximately the target stain space at release, so training-time Macenko introduced a non-linear shift the frozen backbone could not absorb, and the eval-time pass-through cannot recover the small cost paid (−0.022 F1).

The magnitude is modest, even where the lever helps. The +0.020–+0.030 F1 gains on Chaoyang and LungHist700 are real but small; by comparison, swapping the image encoder to PLIP (§4.7) gains +0.027 on Chaoyang and **+0.335 on LungHist700 — an order of magnitude larger**. At the frozen-backbone scale evaluated here, stain normalisation is a useful but not dominant lever, and the protocol lesson worth carrying forward is the asymmetric application: Macenko is applied at inference only on inputs that arrive in their native stain palette, since doubling on a pre-normalised target is what historically broke the OOD F1 in this experiment.

4.4 PLIP zero-shot reference

4.4.1 Configuration

The PLIP zero-shot reference uses PLIP’s Visual Transformer (ViT-B/32) and PLIP’s matching text encoder [7], both frozen and used straight from the model release. No LC25000 supervision is applied: no projection head is trained on our data, no temperature is learned, no checkpoint is fine-tuned. Class prediction proceeds by argmax of cosine similarity between an image embedding and the five LC25000 class-prompt embeddings, both produced by PLIP itself. Evaluation runs on the same LC25000 test split and the same three external datasets as every other variant in this chapter.

This experiment is a **reference point, not an ablation**. It isolates the value of pathology pretraining alone — what an off-the-shelf domain-pretrained model gives on these tasks before any LC25000 supervision is layered on top. The pair-up with the PLIP backbone in §4.7, which keeps PLIP’s image representation but replaces its prompt geometry with a projection head trained on LC25000, becomes the chapter’s clearest decomposition of which pieces of pathology pretraining contribute what.

4.4.2 LC25000 results

On LC25000 the PLIP zero-shot pipeline reaches a macro-F1 of **0.700**. This is the only variant in the chapter where the in-distribution number is informative: every other variant has a projection head trained on LC25000 that drives macro-F1 to ~ 0.99 saturation, whereas PLIP zero-shot has no such supervision, so the gap between 0.700 and saturation directly measures how far an off-the-shelf domain-pretrained model is from a model that has actually seen the training distribution. Within PLIP zero-shot, the per-class F1 spread is wide — 0.87 on benign lung and 0.80 on benign colon, dropping to 0.48 on lung adenocarcinoma and 0.56 on colon adenocarcinoma — so the off-the-shelf model handles healthy/normal tissue patterns more reliably than cancer-subtype distinctions, with no LC25000 supervision to refine the latter.

In the in-distribution UMAP (Figure 4.7) the visual comparison actually favours the baseline: the trained CLIP pipeline produces tighter, more compact class clusters than the off-the-shelf PLIP encoder, because the baseline’s InfoNCE training has explicitly pulled image embeddings toward their matching prompts on LC25000. PLIP zero-shot’s image clusters are looser, with lung adenocarcinoma and lung squamous cell carcinoma bleeding into each other more visibly than in the baseline. PLIP’s pretrained features nonetheless carry enough class-discriminative signal to reach 0.700 macro-F1 with the off-the-shelf classifier alone; the in-distribution UMAP underplays the strength of those features, because their real advantage shows up under shift, not on the same distribution the baseline was trained for.

The visible weakness of the zero-shot reference, in panel (a), is on the text side. In PLIP’s off-the-shelf embedding space the five LC25000 class-prompt embeddings collapse to a near-singular point in the UMAP projection — the single labelled ★ near the centre, with the other four prompt markers overlapping it. Every *trained* variant in this chapter uses its projection head to pull the prompts apart and align each with its image cluster; without that step, the off-the-shelf prompts are too similar to one another to discriminate the five LC25000 classes. The 0.700 in-distribution number reflects this asymmetry: **the pretrained image**

representation carries genuine histopathology signal, but the off-the-shelf prompt classifier built on it caps how far that signal can be exploited.

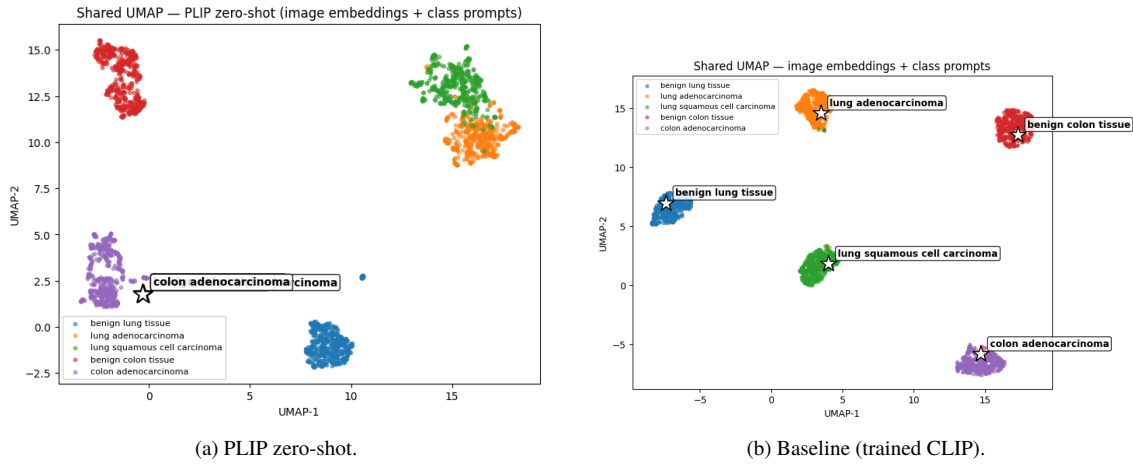


Figure 4.7 Training-source UMAP comparison. Both plots show LC25000 test image embeddings colour-coded by class with class-prompt anchors (★) marked.

4.4.3 Generalization to external datasets

External evaluation is where the off-the-shelf model’s actual strength shows up. Restricted-argmax macro-F1 across the three external datasets is reported in Table 4.4. Two observations dominate. First, PLIP zero-shot is the only variant in the chapter where Chaoyang scores higher than NCT-CRC (0.731 vs 0.661) — every LC25000-trained variant has the opposite ordering. Second, on LungHist700 PLIP zero-shot reaches 0.544 macro-F1 with no LC25000 lung supervision, outperforming every LC25000-trained ResNet50 variant on the same dataset by a large margin (baseline: 0.305).

Table 4.4 PLIP zero-shot generalisation across the three external datasets, restricted-argmax over the target-organ LC25000 classes.

Dataset (organ)	Macro-F1	Accuracy	Bal. acc.	Weighted F1
NCT-CRC-HE-7K (colon)	0.661	0.557	0.610	0.679
Chaoyang (colon)	0.731	0.737	0.761	0.727
LungHist700 (lung, 20×)	0.544	0.621	0.567	0.583

Chaoyang outscoring NCT here is a real effect with a concrete cause in stain distribution, not a noise artefact. NCT-CRC is distributed already Macenko-normalised against an unpublished Kather reference, which deliberately pushes its colour distribution away from the natural variability of typical H&E patches. PLIP’s pretraining corpus (OpenPath, ~ 208k image–text pairs scraped from histopathology Twitter) consists of natively-stained, un-normalised images. Chaoyang, also natively stained at a Beijing hospital, sits closer in colour distribution to what PLIP saw during pretraining; NCT-CRC, pre-normalised at release, sits further away. PLIP’s pretrained image encoder therefore produces better embeddings for Chaoyang’s natural-stain patches than for NCT’s normalised patches — which inverts the colon ordering relative to every LC25000-trained variant, where NCT (closer in style to LC25000’s colon) is the easier external dataset.

The most consequential cell in Table 4.4 is LungHist700: 0.544 macro-F1 from a model that has never seen LC25000 lung supervision, against 0.305 from the LC25000-trained ResNet50 baseline on the same dataset — **a +0.239 F1 gap explained entirely by pathology pretraining vs ImageNet supervision**. The per-class recalls are 0.16 (benign), 0.69 (lung adenocarcinoma), 0.84 (lung squamous cell carcinoma): biased toward squamous like the baseline, but nowhere near the collapse pattern observed for the baseline (0.19/0.07/0.89). Figure 4.8 visualises the cross-distribution structure: PLIP-encoded LungHist700 patches sit close to LC25000’s lung clusters in a single shared embedding space, where the baseline produced no usable lung structure at all. This is the chapter’s first concrete evidence that the image-encoder lever is

the dominant one, and §4.7 returns to it by replacing PLIP’s off-the-shelf prompt geometry with a trained projection head on LC25000.

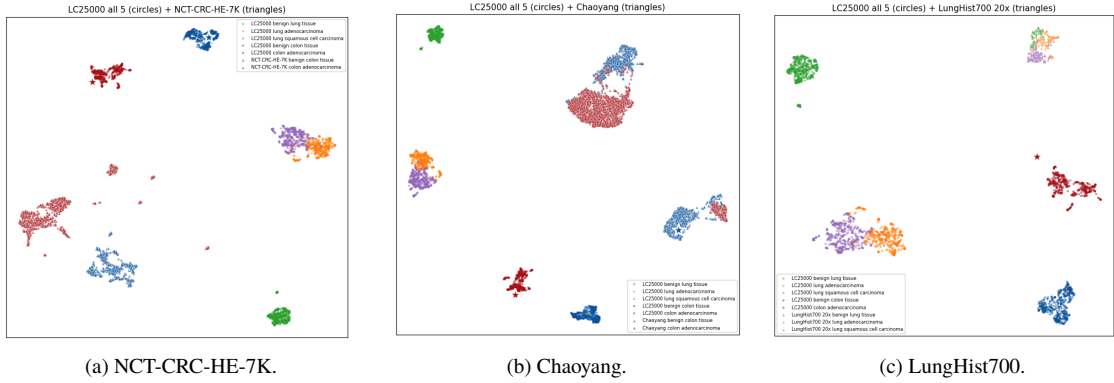


Figure 4.8 PLIP zero-shot cross-distribution UMAPs across the three external datasets. Each panel shows LC25000 image embeddings (dots, colour-coded by class) overlaid with the external dataset’s images (triangles), projected into the same 2-D fit per dataset.

4.5 Foundational model vs plain classifier

4.5.1 Configuration

The plain-classifier comparison strips the CLIP paradigm from the baseline pipeline while holding the image side untouched. The frozen ResNet50 image encoder from §4.2.1 is reused verbatim — same checkpoint, same preprocessing, same 80/10/10 split on LC25000 — and everything CLIP-specific is removed: no text encoder, no projection heads, no learnable temperature, no contrastive loss. In its place, a non-CLIP closed-set head is layered on top: ResNet50 features pass through global average pooling, a single BatchNorm layer, and a Dense(5) output trained with sparse categorical cross-entropy. The optimiser, learning rate, and training budget mirror the baseline.

The point of this configuration is to isolate the CLIP paradigm itself as the experimental variable. Image encoder, training data, and adaptation budget are all held constant; the only thing that changes is the classification head — a contrastive image-text matching against five prompts on one side, a softmax over five direct logits on the other. Any external-evaluation gap between the two therefore measures what the CLIP paradigm contributes once the underlying image representation is the same. This is the RQ2 test: does the foundational-model framing earn its complexity, or could a plain closed-set classifier match it under the same conditions?

4.5.2 Generalization to external datasets

The two heads tie on the training distribution. Macro-F1 on the LC25000 test split is 0.996 for the baseline and 0.995 for the plain classifier — both saturated at the ceiling that any variant with a head trained on LC25000 reaches in this chapter. The story is in the external columns of Table 4.5. Plain cross-entropy is narrowly ahead on NCT-CRC (−0.014 F1 for CLIP), behind by a meaningful margin on Chaoyang (+0.064 F1 for CLIP), and behind by a modest margin on LungHist700 (+0.024 F1 for CLIP). **The CLIP paradigm’s edge is invisible on the training distribution, narrowly negative on the closest external dataset, and grows with shift severity from there.**

The Chaoyang gap is not a uniform regression — it has a concrete collapse signature in the per-class metrics. Plain CE reaches benign-colon precision 0.96 but recall only 0.47: the classifier is rarely predicting benign at all, but is right when it does. Its adenocarcinoma precision is 0.70 with recall 0.98 — almost every patch is being routed to the cancer class. This is a near-degenerate “predict adenocarcinoma” regime on a 2-class dataset. The baseline CLIP variant on the same images stays balanced (benign 0.86/0.65, adeno 0.76/0.92): imperfect, but not collapsed. Same image features, same supervision, same evaluation — only the classifier head differs.

Table 4.5 CLIP (baseline) vs plain classifier on the same frozen ResNet50 image encoder, macro-F1 across the training-source test split and three external datasets.

Variant	LC25000	NCT	Chaoyang	LungHist700
Baseline (CLIP InfoNCE)	0.996	0.866	0.786	0.305
Plain classifier (CE)	0.995	0.880	0.722	0.281
Δ (CLIP – plain)	−0.001	−0.014	+0.064	+0.024

The collapse comes from the classifier head, not from the image features themselves. The plain head learns one score per class and predicts the class with the highest score; training on LC25000 tunes those scores to be sharp, so the winning class beats the others by a wide margin on LC25000 patches. Under shift the learned scores stay where training left them, but the image features now sit in a different region of the embedding space — and there one class often scores highest for nearly every patch, regardless of true label. The sharp softmax then routes almost all the probability mass to that single class. That is the single-class collapse seen on Chaoyang. CLIP’s head, by contrast, scores cosine similarity between an L2-normalised 256-d image projection and a 256-d prompt embedding, scaled by a learned temperature τ . Three properties of this geometry stabilise behaviour under shift:

- Cosine similarity is bounded in $[-1, 1]$, removing the unbounded-margin room a softmax has to over-extrapolate.
- The L2-normalised space concentrates representations near the unit sphere, so samples that drift in absolute feature space remain closer in *angular* terms to the prompt anchors.
- The learned temperature τ , optimised end-to-end on training data, tempers extreme confidence without collapsing the ranking signal.

The plain CE head has none of these buffers.

This is a property of the CLIP paradigm rather than of any specific encoder choice. CLIP’s head geometry was designed for the open-vocabulary regime, where new class prompts can be introduced at inference time without retraining; the same geometric choices — bounded cosine, normalised features, learned temperature — also turn out to dampen confidence drift under distribution shift on closed-set tasks like ours. The result validates RQ2’s foundational-model framing in a specific way: **CLIP does not earn its complexity at peak in-distribution accuracy — on LC25000 a plain classifier is indistinguishable — but in robustness under shift**, with margins that scale with how far the external dataset sits from the training source. The open-vocabulary capabilities (retrieval, prompt-tuning at inference, zero-shot to unseen classes) are not exercised by the closed-set evaluation in this chapter and remain a parallel argument for the paradigm.

4.6 Text encoder pretraining

4.6.1 Configuration

The text-encoder comparison keeps the entire baseline pipeline from §4.2.1 frozen and changes only one component: the text model. The image encoder, both projection heads, the InfoNCE loss, the LC25000 split, and the AdamW training schedule are all reused unchanged. Three text encoders are compared — `bert-base-uncased` (the baseline’s general-purpose checkpoint), BioBERT, and PubMedBERT — all BERT-family, all producing 768-d text projections, all using wordpiece tokenisation. The architecture is therefore identical across the three; the only thing that differs is the text corpus each encoder was pretrained on.

On top of the encoder swap, two prompt formats are tested: **name_only**, where the prompt is just the bare class name (e.g. ‘colon adenocarcinoma’), and **composed**, the baseline’s template ‘A histopathology image of {class}.’. The cross match gives a $3 \times 2 = 6$ -cell grid. The point of testing prompt length is that biomedical pretraining mostly affects how the model handles medical vocabulary — and that vocabulary lives in the class name itself, not in the surrounding boilerplate. If biomedical encoders carry an advantage at all, it should be sharpest when the prompt strips down to the class name. This is the RQ4 test: does swapping in a more domain-aware text encoder buy anything for histopathology classification, and does the prompt format change the answer?

4.6.2 NCT-CRC grid (6 cells)

The first sweep crosses all six cells against the closest external dataset, NCT-CRC. The full picture is in Table 4.6: LC25000 in-distribution saturates at ~ 0.99 for every cell (the LC25000 axis does not discriminate these encoders), and the NCT macro-F1 scores all land in a narrow 0.81–0.86 band. The baseline (BERT + composed, equivalent to the §4.2.1 configuration) sits at 0.866. The best *biomedical* cell — PubMedBERT + name_only — reaches 0.855, still 0.011 below baseline. **None of the six cells beats the baseline on NCT.** The UMAP grid in Figure ?? confirms what the numbers say: cluster geometry looks qualitatively similar across all six cells, so the projection head is producing comparable class anchors regardless of the starting text encoder.

Table 4.6 6-cell biomedical text encoder ablation, macro-F1 on NCT-CRC. The BERT + composed cell is essentially baseline (same architecture, same prompt template); the best *biomedical* cell — PubMedBERT + name_only at 0.855 — still trails baseline by -0.011 F1.

Cell	LC25000	NCT-CRC	Δ vs baseline NCT
BERT + name_only	0.999	0.821	-0.045
BERT + composed	0.999	0.863	-0.003
BioBERT + name_only	0.998	0.840	-0.026
BioBERT + composed	0.998	0.818	-0.048
PubMedBERT + name_only	0.995	0.855	-0.011
PubMedBERT + composed	0.996	0.808	-0.058

Three converging reasons explain why biomedical text pretraining doesn’t translate into a measurable advantage here. First, the projection head absorbs encoder-specific differences. BERT, BioBERT, and PubMedBERT all start from different 768-d embedding spaces, but all three get pulled through the same trainable projection head onto LC25000’s five class anchors — with only five classes to separate, that head learns to map any reasonable starting embedding onto useful anchors, washing out the encoder differences in the process. Second, short class-name prompts don’t expose biomedical-vocabulary richness. Strings like “colon adenocarcinoma” tokenise into 2–3 wordpieces that any BERT-family model handles competently; biomedical pretraining adds value most where domain vocabulary is dense (clinical notes, paper abstracts, multi-sentence text), and the prompts here aren’t dense at all. Third, **CLIP’s robustness under shift is geometric, not knowledge-driven** — the cosine + L2-norm + temperature classifier established in §4.5 is independent of which text encoder produces the prompt embeddings. A different encoder just moves the starting positions of the anchors; the projection head re-maps them anyway, and the geometry that protects against shift stays the same.

4.6.3 Generalization to Chaoyang and LungHist700

Even though no biomedical cell beats baseline on NCT, it could still be that biomedical pretraining pays off under heavier shift. To check this, the best biomedical cell on NCT — PubMedBERT + name_only — was extended to both Chaoyang and LungHist700 and compared against the baseline (BERT + composed). Table 4.7 shows the result: PubMedBERT + name_only ties baseline on Chaoyang within noise ($+0.009$) and regresses on LungHist700 (-0.026), in addition to its -0.011 on NCT. **No measurable transfer benefit on any of the three external datasets.**

Table 4.7 Best biomedical text-encoder cell (PubMedBERT + name_only) vs baseline on the three external evaluation datasets. The biomedical text encoder ties baseline on Chaoyang (within noise) and regresses on both NCT and LungHist700. No measurable transfer benefit on any of the three datasets.

External dataset	Baseline F1	PubMedBERT+name_only F1	Delta
NCT-CRC-HE-7K (close-shift colon)	0.866	0.855	-0.011
Chaoyang (different-hospital colon)	0.786	0.795	$+0.009$
LungHist700 (different-source lung, $20\times$)	0.305	0.279	-0.026

On LungHist700, both variants collapse to a single class — and the collapse class is the same. Baseline routes most of its predictions to “lung squamous cell carcinoma” (the $\sim 84\%$ behaviour documented in §4.2.3); PubMedBERT + name_only collapses to the same class, only harder ($\sim 92\%$ of predictions). In both cases per-class recall on the other two lung classes (benign, adenocarcinoma) is near-zero. The text-encoder choice does not change the direction of the collapse — only how strongly the model commits to that single class.

One secondary observation in Table 4.6 is worth flagging because it might be useful for anyone designing similar ablations. Vanilla BERT prefers the longer *composed* prompt (0.863 vs 0.821 on NCT), while both biomedical encoders prefer the bare *name_only* format (PubMedBERT: 0.855 vs 0.808; BioBERT: 0.840 vs 0.818). The likely explanation is that biomedical pretraining made strings like “colon adenocarcinoma” carry strong signal on their own, so the generic “A histopathology image of ...” boilerplate just dilutes that signal — whereas vanilla BERT benefits from the surrounding context to lock in a coherent embedding. None of this flips any cell to winning against baseline, but it’s a useful note for short-prompt design.

4.7 Image encoder pretraining

4.7.1 Configuration

The image-encoder comparison keeps the architecture, loss, prompts, train/val/test split, and frozen BERT text encoder from §4.2.1 all unchanged. The only axis that moves is the image encoder’s pretraining. Three encoders are compared head-to-head, all frozen — the only trainable parameters in each variant are the projection head on top and the learned temperature τ .

- **ResNet50 (ImageNet, supervised)** — the baseline’s CNN, trained with class labels on natural images.
- **ViT-B/14 (DINOv2, self-supervised)** — a Visual Transformer pretrained on $\sim 142\text{M}$ curated general natural images via self-distillation, no labels, no histopathology.
- **ViT-B/32 (PLIP / OpenPath, contrastive)** — a Visual Transformer pretrained on $\sim 208\text{k}$ histopathology image–text pairs scraped from Twitter via a CLIP-style contrastive objective.

DINOv2 is the experimental control for this comparison. Without it, swapping ResNet50 for PLIP changes three things at once: the architecture (CNN $\rightarrow$ ViT), the pretraining objective (supervised classification $\rightarrow$ image–text contrastive), and the pretraining domain (natural images $\rightarrow$ histopathology). Any improvement could be attributed to any of the three. DINOv2 fixes this by holding architecture and the broad “general-domain SSL” axis constant while leaving only the domain different from PLIP. Concretely, if DINOv2 also breaks the baseline’s lung collapse, part of PLIP’s advantage is architectural rather than domain-specific; if DINOv2 stays at baseline-level performance, the advantage is specifically about histopathology pretraining.

One known confound is worth flagging upfront. PLIP uses ViT-B/32 (patch size 32×32) while DINOv2 uses ViT-B/14 (patch size 14×14). At the same 224×224 input, that gives DINOv2 256 patch tokens to PLIP’s 49 — markedly different feature granularity. We track this as a potential confound throughout the section: if a result depends specifically on PLIP’s pretraining, we should see DINOv2 fail to match it, despite DINOv2’s higher spatial resolution. The patch-size axis is documented as a limitation in Chapter 5.

4.7.2 Generalization to external datasets

On LC25000 the three variants are indistinguishable: 0.996 for PLIP, 0.996 for the ResNet50 baseline, 0.989 for DINOv2 — all three saturate at the dataset’s ceiling. Notably, DINOv2 saturates too despite having never seen labelled histology during pretraining: the LC25000 axis cannot tell these encoders apart. The story is in the external columns of Table 4.8. **PLIP wins on all three external datasets** (+0.067 on NCT, +0.027 on Chaoyang, +0.335 on LungHist700). DINOv2 is more selective — $-0.011 / -0.010$ on the two colon datasets (within noise of baseline) but +0.152 on LungHist700. The lung gap is by far the largest single delta anywhere in this chapter; the rest of this section explains where it comes from.

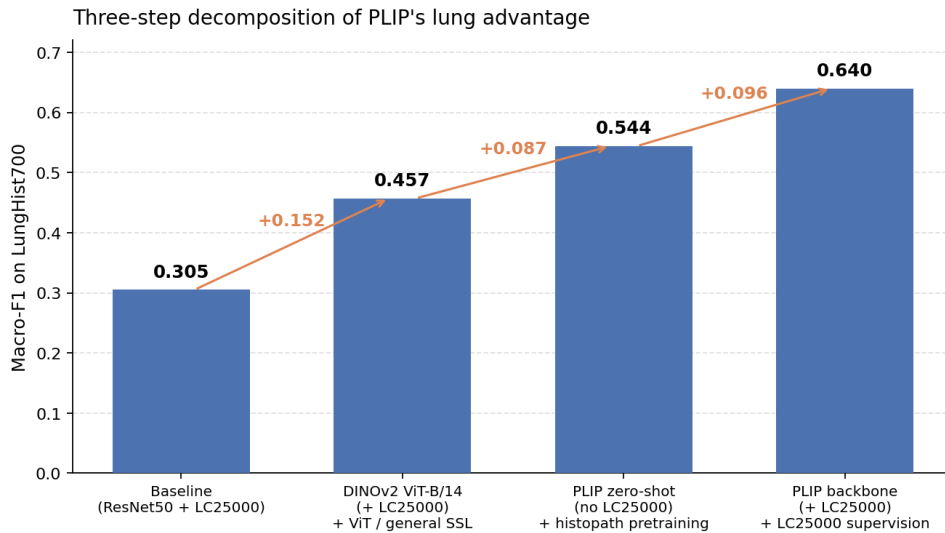
Starting on the colon side. On NCT-CRC PLIP gains +0.067 F1 over the baseline and on Chaoyang +0.027 F1 — meaningful improvements, but in the same order of magnitude as input-side levers elsewhere in the chapter (stain Macenko: +0.030 on Chaoyang). DINOv2 on the same two datasets lands at -0.011

Table 4.8 Image encoder pretraining swap, macro-F1 across the training-source split and three external datasets. Everything except the image encoder is identical across rows.

Image encoder	LC25000	NCT	Chaoyang	LungHist700
ResNet50 (ImageNet)	0.996	0.866	0.786	0.305
ViT-B/14 (DINOv2 / self-supervised)	0.989	0.855	0.776	0.457
ViT-B/32 (PLIP / OpenPath)	0.996	0.933	0.813	0.640

and -0.010 , both within noise of baseline. **On colon, the advantage is specifically about the pretraining domain, not the architecture or the self-supervised paradigm:** PLIP’s gain is what histopathology pretraining buys; DINOv2’s general-domain SSL on $\sim 142\text{M}$ natural images doesn’t help because the ResNet50 baseline was already extracting features adequate for partial colon transfer to NCT and Chaoyang. The bottleneck on colon was never the architecture’s feature breadth.

The lung side is qualitatively different, and the DINOv2 control plus the PLIP zero-shot reference from §4.4 together decompose PLIP’s full $+0.335$ F1 gap into a clean three-step ladder (Figure 4.9). The baseline reaches 0.305 macro-F1 on LungHist700 — effectively a degenerate single-class predictor. Swapping in DINOv2’s ViT backbone (still no histology pretraining) lifts that to 0.457, a $+0.152$ gain attributable to “ViT + general-domain SSL”. Replacing DINOv2’s general-domain pretraining with PLIP’s histopathology-specific pretraining — but with no LC25000 supervision, the zero-shot variant — reaches 0.544, an additional $+0.087$ gain attributable specifically to “histopathology pretraining”. Finally, layering LC25000 supervision on top of PLIP’s features brings it to 0.640, an additional $+0.096$ gain attributable to “task supervision on top of histopathology pretraining”. **Three roughly equal ~ 0.1 F1 steps**, sequentially adding architecture/SSL, then domain, then task supervision.

**Figure 4.9** LungHist700 macro-F1 across the four variants, with the contribution added at each step labelled.

The per-class story makes the same point concretely. The baseline collapses to predicting “lung squamous cell carcinoma” on $\sim 84\%$ of inputs (per-class recall: 0.19 / 0.07 / 0.89 for normal / adenocarcinoma / SqC), and a plain CE classifier on the same ResNet50 features collapses similarly (0.15 / 0.05 / 0.91). DINOv2 breaks the collapse and produces a balanced predictor with recalls 0.46 / 0.52 / 0.40 — balanced but with a lower peak. PLIP combines both properties: balanced *and* high peak, with recalls 0.51 / 0.49 / 0.91. Figure 4.10 visualises the same story in feature space: ResNet50 places all 359 LungHist700 samples into a single tight cluster sitting next to LC25000’s colon adenocarcinoma region (mis-located, no internal lung structure); DINOv2 partially relocates them into a lung-region cluster with class overlap; PLIP separates them into three visibly-distinct sub-clusters by class, sitting adjacent to the matching LC25000 lung clusters.

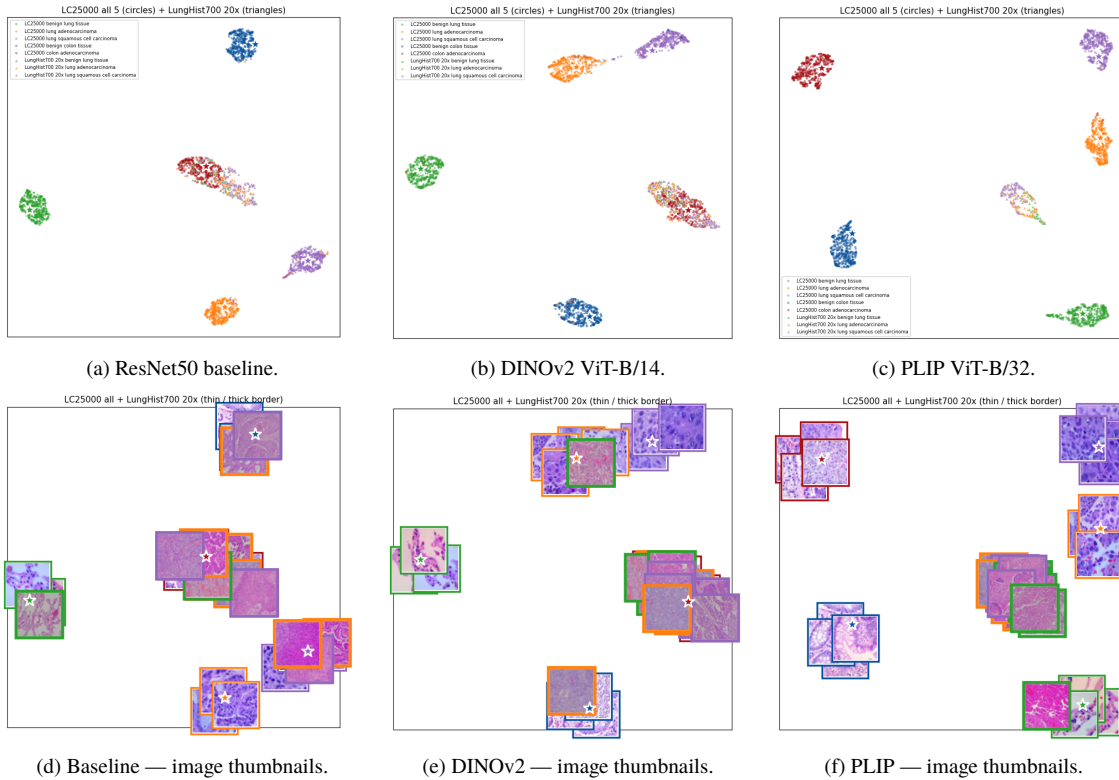


Figure 4.10 LungHist700 cross-distribution UMAP across the three image encoders. Top row: scatter views with LC25000 classes (circles) and LungHist700 (triangles) overlaid in a shared 2-D UMAP fit per encoder. Bottom row: the same UMAP positions with representative image thumbnails overlaid; thin borders mark LC25000 patches, thick borders mark LungHist700 patches, and border colour encodes class.

Why does the encoder swap help so much more on lung than on colon? The ResNet50 baseline was pretrained on ImageNet — supervised classification on natural images, with no histopathology. Whatever colon signal it carries to external data comes from LC25000 supervision finding ImageNet features that happen to correlate with the colon labels in LC25000 itself — a correlation that transfers partially to NCT and Chaoyang (both colon, stylistically close) but not to LungHist700, where the in-distribution lung accuracy is essentially memorisation on the LC25000 lung subset. DINOv2’s self-supervised pretraining on $\sim 142\text{M}$ diverse natural images produces image features sensitive to a much wider range of local-texture variation than ImageNet supervision does, and that broader feature space is enough for LC25000 supervision plus the projection head to assemble a usable lung classifier — even though DINOv2 itself has never seen a histology image. On colon, where the baseline’s features were already adequate, DINOv2’s extra breadth has nothing left to add.

4.8 Discussion

4.8.1 Synthesis across experiments

Looking back across the chapter, the experiments fall into two camps. Most of what was tested — stain Macenko normalisation, biomedical text encoders, prompt-format variation, and a plain closed-set classifier head — produces changes within ~ 0.03 F1 of the baseline on the closest external dataset (NCT-CRC) and at most ~ 0.03 F1 of improvement on the harder external datasets. Stain Macenko is the only one of these input-side or head-side interventions that produces a positive gain under shift ($+0.030$ on Chaoyang, $+0.020$ on LungHist700, but -0.022 on NCT because NCT arrives already Macenko-normalised). The best biomedical text-encoder cell (PubMedBERT + name_only) ties baseline on Chaoyang and regresses on both NCT and LungHist700 — no measurable transfer benefit on any of the three external datasets.

The image-encoder swap is the exception. Replacing the ResNet50 baseline’s pretraining moves the headline metric by +0.067 on NCT, +0.027 on Chaoyang, and +0.335 on LungHist700 — the last one taking the lung result from a degenerate single-class predictor at 0.305 to a functional 3-class classifier at 0.640. The DINOv2 and PLIP zero-shot controls decompose that lung gain into a clean three-step ladder ($0.305 \rightarrow 0.457 \rightarrow 0.544 \rightarrow 0.640$), each step adding one of (architecture + general-domain SSL), (histopathology pretraining), or (LC25000 supervision on top), with the three contributions roughly equal ~ 0.1 F1 each. **On LungHist700 the image-encoder lever is decisive — the only intervention in this chapter that moves a non-functional predictor into a functional one.** On NCT it is the largest single positive lever; on Chaoyang it effectively ties stain Macenko (+0.027 vs +0.030).

4.8.2 Secondary observations

Cross-organ regime change. ResNet50 variants partially work on colon external data (macro-F1 0.78–0.88) but categorically fail on lung (around 0.30, near 3-class chance). What’s happening here is that LC25000 supervision can rescue ResNet50 features that happen to correlate with colon class labels in LC25000, but it cannot manufacture useful lung-tissue features inside a frozen ImageNet-pretrained ResNet50; the in-distribution lung accuracy is essentially memorisation. **Per-organ external evaluation is therefore essential:** aggregating across datasets would hide the regime change.

Classification quality is not the same as representation alignment. The baseline CLIP variant achieves cross-dataset *cluster* alignment in feature space but lower classification F1; PLIP achieves the highest F1 but leaves external samples in their own region of feature space, not co-located with the LC25000 clusters. The contrastive objective optimises same-class-pair alignment; the cosine-classifier rule only needs alignment to the class-prompt direction. These are different geometric properties — the right tradeoff between them depends on whether the downstream task is classification or representation-driven (retrieval, clustering, transfer).

Plain-classifier flip under shift. The plain CE classifier ties CLIP in-distribution and marginally wins on close-shift NCT, but loses Chaoyang (+0.064 F1 for CLIP) and LungHist700 (+0.024 F1 for CLIP). As §4.5 established, the reason is that a high-confidence softmax classifier degenerates into a single-class predictor as shift severity grows, while CLIP’s bounded-cosine geometry does not. **This supports RQ2’s foundational-model framing: CLIP earns its complexity in shift robustness, not in-distribution accuracy.**

4.8.3 Headline summary

Table 4.9 gathers the per-dataset macro-F1 for every variant evaluated in this chapter. **The headline finding is that the PLIP image encoder wins every external evaluation, and on LungHist700 it does so by an order of magnitude more than any other lever tested** (+0.335 F1 over baseline vs at most +0.030 for any other intervention). Chapter 5 returns to the limitations and future work.

Table 4.9 Macro-F1 across all variants evaluated in this chapter, on LC25000 (training source) and the three external datasets.

Variant	LC25000	NCT	Chaoyang	LungHist700
ViT-B/32 (PLIP)	0.996	0.933	0.813	0.640
ViT-B/14 (DINOv2)	0.989	0.855	0.776	0.457
Baseline (ResNet50 + BERT)	0.996	0.866	0.786	0.305
Plain CE classifier	0.995	0.880	0.722	0.281
Stain Macenko (NCT-NORM reference)	0.986	0.844	0.816	0.325
Best biomedical text-encoder cell (PubMedBERT + name_only)	0.995	0.855	0.795	0.279
PLIP zero-shot (no LC25000 supervision)	0.700	0.661	0.731	0.544

5 Conclusions and Future Work

5.1 Conclusions

Headline. The image encoder’s pretraining is the dominant lever in this study. Holding everything else fixed, swapping the general-purpose ResNet50 (ImageNet supervised) for PLIP’s pretrained ViT-B/32 (histopathology image–text contrastive) improves external evaluation by +0.067 F1 on close-shift colon (NCT-CRC), +0.027 on different-hospital colon (Chaoyang), and +0.335 on different-source lung (LungHist700). None of the other five interventions tested moves the needle comparably. The DINOv2 and PLIP zero-shot controls together decompose PLIP’s lung advantage into three roughly equal ~ 0.1 F1 steps: ViT + self-supervised pretraining (+0.152), histopathology pretraining specifically (+0.087), and LC25000 supervision on top (+0.096).

RQ1 — Baseline behavior and failure modes

The frozen general-purpose CLIP pipeline (ResNet50 + BERT) saturates LC25000 at ~ 0.99 macro-F1, transfers partially to colon external data (0.866 on NCT-CRC, 0.786 on Chaoyang), and categorically fails on lung external data (0.305 on LungHist700, near 3-class chance). The lung failure is a regime change rather than a smooth degradation: both ResNet50-based variants collapse into a single-class predictor, routing $\sim 90\%$ of inputs to “lung squamous cell carcinoma” regardless of true class. See Results §4.2.

RQ2 — CLIP’s foundational-model premise

CLIP’s foundational-model premise is validated under shift, not in-distribution. On LC25000 CLIP and a plain cross-entropy classifier on the same frozen ResNet50 backbone tie at saturation, and on close-shift NCT-CRC they tie marginally with plain CE narrowly ahead (+0.014 F1). CLIP’s edge appears only under heavier shift: +0.064 F1 on Chaoyang and +0.024 F1 on LungHist700. This comes down to classifier geometry: CLIP’s bounded-cosine, L2-normalised, temperature-scaled head is more robust than a high-confidence softmax, which under shift degenerates into a single-class predictor (visible on Chaoyang and LungHist700 in the per-class signatures). See Results §4.5.

RQ3 — Stain normalization and generalization

Stain normalization’s direction is data-dependent. Macenko (with the NCT-NORM reference) applied at training, and the same transform applied at inference only on external datasets that arrive in their native stain palette, helps on Chaoyang (+0.030 F1) and LungHist700 (+0.020 F1) and regresses on NCT-CRC (−0.022 F1, which is already Macenko-normalised at release). The story is intuitive in both directions: when an external input genuinely sits in a different stain palette than the training distribution, applying Macenko at inference moves it closer to the model’s training-stain space and the frozen backbone produces better features. On NCT, where the input was already approximately in the target stain space, training-time Macenko added a non-linear input-statistic shift that the frozen backbone could not absorb, and there was no gain at evaluation to compensate.

Even where Macenko helps, the magnitude is modest. The Chaoyang gain (+0.030) is comparable to the image-encoder lever (+0.027 for PLIP), and on LungHist700 it is an order of magnitude smaller (+0.020 for Macenko vs +0.335 for PLIP). Stain normalization is a useful but not dominant lever. See Results §4.3.

RQ4 — Biomedical text encoder pretraining

Biomedical text-encoder pretraining produces no measurable transfer benefit at this scale. None of the six cells (BERT / BioBERT / PubMedBERT × bare-name / composed-prompt) beats the baseline on NCT external evaluation; all land in a narrow 0.81–0.86 macro-F1 band. Extending the best biomedical cell (PubMedBERT + name_only) to Chaoyang and LungHist700 tells the same story — it ties baseline on Chaoyang (+0.009, within noise) and regresses on NCT (−0.011) and LungHist700 (−0.026). On LungHist700 the model still collapses to a single-class predictor (~92% squamous), the same failure mode as the baseline, only sharper. Three reasons stack up here: the trained projection head absorbs encoder-specific differences in starting embedding space; short class-name prompts do not expose biomedical-vocabulary richness; and CLIP’s shift robustness is a property of the classifier geometry, not of the text encoder’s domain knowledge.

A small secondary observation is worth keeping for anyone designing similar experiments: vanilla BERT prefers the longer *composed* prompt while biomedical encoders prefer the bare *name_only* format. None of this flips any cell to winning, but it’s a useful note for short-prompt design. See Results §4.6.

RQ5 — Image encoder pretraining (the dominant lever)

The image encoder’s pretraining is the dominant lever in this study. Replacing ResNet50 with PLIP’s pretrained ViT-B/32 (histopathology image–text contrastive) gains +0.067 F1 on NCT, +0.027 on Chaoyang, and +0.335 on LungHist700. None of the head, loss, or text-side interventions tested gets close. The DINOv2 control plus the PLIP zero-shot reference decompose the lung gain into a clean three-step ladder (0.305 → 0.457 → 0.544 → 0.640), each step adding one of (ViT + general-domain SSL, +0.152), (histopathology pretraining, +0.087), or (LC25000 supervision on top, +0.096), with the three contributions roughly equal ~0.1 F1 each. The colon side decomposes differently: only histopathology pretraining moves the needle there, and general-domain SSL ties baseline within noise. The lever’s contribution scales with shift severity.

Set against the broader CLIP-for-histopathology literature, this is a counterintuitive finding. Recent papers routinely emphasise novel contrastive losses, biomedical text encoders, and stain harmonisation as the key levers for cross-dataset generalisation. At the frozen-backbone scale evaluated here, this study does not find support for that emphasis — only the image-encoder swap moves the headline metric meaningfully, and it does so most decisively on the hardest dataset (LungHist700). See Results §4.7.

RQ6 — Generalization across 3 external datasets and 2 organs

PLIP’s ViT-B/32 wins every external evaluation in the study (NCT 0.933, Chaoyang 0.813, LungHist700 0.640), and its lung-side advantage over the baseline (+0.335 F1) is an order of magnitude larger than its colon-side advantages (+0.067 on NCT, +0.027 on Chaoyang). The cross-organ asymmetry is itself a regime change: ResNet50-based variants transfer partially to colon (0.78–0.88 F1) but categorically fail on lung (~0.30, near 3-class chance), while PLIP shows no such asymmetry. **Per-organ external reporting is therefore essential:** aggregating across datasets would have hidden the regime change.

What unlocks cross-organ transfer is feature breadth, not specifically histopathology knowledge. The DINOv2 control demonstrates this directly: DINOv2’s self-supervised pretraining on ~142M diverse natural images contains no histopathology at all, yet it lifts LungHist700 from 0.305 to 0.457 and produces a balanced 3-class predictor where the ImageNet-supervised ResNet50 collapsed. Narrow ImageNet-supervised features are too brittle for cross-organ transfer; either massive general-domain SSL or domain-specific (histopathology) pretraining is enough to unlock it. Histopathology pretraining additionally amplifies the gain — both on colon (where general-domain SSL ties baseline) and on lung (where it stacks on top of the SSL contribution) — but it is not strictly required to break the cross-organ collapse. See Results §4.7 and §4.8.

Cross-cutting takeaways

Two observations cut across multiple RQs and deserve calling out. First, **LC25000 accuracy says nothing useful about how a variant behaves under shift**. Every trained variant we tested saturates LC25000 at ~ 0.99 macro-F1, while external macro-F1 across the same variants ranges from 0.28 to 0.93. A reader looking only at the training-source numbers would have called nearly all of our variants identical — when in fact they differ by up to 0.65 F1 on external data. The practical consequence: in this kind of saturated frozen-backbone setup, you need external evaluation to distinguish variants at all.

Second, **CLIP’s robustness under shift is geometric, not knowledge-based**. RQ2 establishes this directly: the same frozen image features, classified by CLIP’s cosine head, beat the same features classified by a softmax head under shift. RQ4 confirms it from the other direction: swapping in a biomedical text encoder — adding more medical domain knowledge to the text tower — does not produce a measurable transfer benefit. The cosine + L2 + temperature geometry is what does the work; the components themselves can be domain-agnostic.

5.2 Limitations

The data side has three caveats. LC25000 is a single-institution release with no patient-level grouping or clinical validation, and although the three external datasets used here (NCT-CRC, Chaoyang, LungHist700) span two organs and a meaningful range of shift severity, each is itself a single-institution release. Two LC25000-internal split artefacts affect in-distribution but not external numbers: 1 250 source images $\times$ 20 augmentations are scattered randomly across the 80/10/10 split (likely the dominant inflator of the in-distribution F1 ceiling), and the canonical pre-dedupe split additionally lets 1 280 byte-identical duplicates cross train/test (effect ~ 1 – 2 F1 points). External evaluations are unaffected by both.

A few methodological choices set the boundary of what this study can conclude. Every variant we evaluated keeps the image backbone frozen and only trains a projection head and temperature on top. We didn’t fine-tune end-to-end — this is the regime where a histopathology foundation model is typically used, but if you unfroze the backbone, which lever dominates could easily shift. We also ran each condition with a single random seed, so we don’t have a variance estimate per cell. That matters most for LungHist700 (only 359 images split 85/146/128 across the three classes); putting bootstrap confidence intervals around the headline lung gap is the first item on our future-work list. Finally, external evaluation uses restricted argmax over the LC25000-mappable classes (2-class colon for NCT and Chaoyang, 3-class lung for LungHist700) — richer multi-class scenarios on those datasets weren’t tested here.

A handful of per-experiment caveats round out the picture. The image-encoder comparison mixes two things at once: PLIP uses ViT-B/32 and DINOv2 uses ViT-B/14, so at 224×224 input they see 49 vs 256 patch tokens. DINOv2’s smaller patches do not rescue the colon-side gap, which suggests patch size isn’t the dominant factor — but to fully isolate “pretraining domain” as a variable, the two ViTs would need matching patch sizes. The stain-normalization experiment uses a single NCT-NORM reference patch for every external evaluation; picking a different reference could move the numbers. LungHist700’s source bias hasn’t been characterised, so part of the ResNet50 lung collapse could be a stain artefact on top of the architectural limit, not just architecture. And on the comparison side, we ran no human-expert or clinical validation, and no head-to-head against other histopathology foundation models (UNI, CONCH, Virchow-2, BiomedCLIP, MUSK, PathChat). That last one would have told us whether PLIP specifically wins, or whether any histopathology-pretrained backbone would — it’s purely a compute-and-time call, and it’s on the longer-term future-work list.

5.3 Future Work

Medium-term

- Push the InfoNCE training to a larger effective batch (via gradient accumulation or a larger GPU) and check whether the shared-prompt false-negative regime documented in §3.3 is actually binding here, or whether the temperature behaviour we observed would survive at a larger N .

- Try prompt-tuning methods (CoOp [28], CoCoOp) on top of the frozen encoders — a learned prompt might close some of the colon-side gap without touching the backbone.

Longer-term

- Controlled head-to-head against other histopathology foundation models: UNI [21], CONCH [20], BiomedCLIP [22], MUSK [23]. The current study tells us that PLIP wins under our specific setup, but not whether any histopathology-pretrained backbone would win equally well, or whether PLIP is doing something the others are not.
- Multi-institution training and evaluation — the natural fix for the single-institution limitation flagged above.
- **Alternative encodings for whole-slide images.** Patch-level approaches like the ones in this thesis are practical but throw away a lot of slide-level context. One direction worth exploring is treating a WSI as a sequence of tiles (a “time series”) and training a multimodal model that ingests this sequence alongside accompanying text — a fundamentally different way of encoding the gigapixel input.
- **Self-supervised pretraining on The Cancer Genome Atlas (TCGA).** TCGA provides a large corpus of histopathology data without dense semantic labels — a natural fit for self-supervised pretraining at scale before downstream classification or segmentation. This would build a histopathology-domain backbone under controlled conditions, complementing the off-the-shelf PLIP / CONCH / UNI backbones tested here.
- Returning to dense prediction with pathology foundation-model backbones — the segmentation question that motivated the early framing of this thesis.
- Applying CLIP-with-short-labels to other low-label scientific domains.

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